

# THE EMERGENCE OF COMPLEX BEHAVIOR IN LARGE-SCALE ECOLOGICAL ENVIRONMENTS

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## ABSTRACT

We investigate how physical scale and population size parameterize the emergence of complex behavior in open-ended ecological environments. In the proposed setting, agents are not supervised and receive neither explicit rewards nor externally specified learning objectives; instead, policy populations change over time through reproduction, mutation, and selection. Agent actions simultaneously modify local environmental state and the surrounding population, producing a coupled ecological dynamical system. The objective of this study is therefore not to optimize a single high-performing policy, but to characterize how behavioral repertoires arise and persist in large populations under natural competition and environmental constraint. Using modern hardware and a new multi-agent simulator, we scale both environment size and population size beyond prior configurations, reaching populations of more than 60,000 agents, each parameterized by its own evolved neural network policy. We identify several emergent behavioral classes, including long-range resource extraction, vision-mediated foraging, and predation, that arise under competitive and survival pressures. We further analyze how sensor modality and environmental scale affect the appearance of these behaviors, finding that some behaviors are observed only when environments and populations are sufficiently large, and that increased scale improves the stability and cross-seed consistency of the resulting dynamics. Although evolutionary computation has a substantial research history, our scaling results on contemporary hardware indicate a promising technical direction for using ecology as an instrument for machine learning under increasingly abundant computational resources and efficient machine-learning frameworks. Experimental code will be released upon acceptance.

## 1 INTRODUCTION

Since Darwin, the evolutionary emergence of biodiversity and complex behavior has been analyzed across multiple biological disciplines, including evolutionary theory (Darwin 1859, Simpson 1944), ecology (MacArthur & Wilson 2001), and genetics (Dobzhansky 1937). Formal and empirical models of these dynamics range from accounts of speciation and adaptive radiation (Coyne & Orr 2004, Schluter 2000) to mathematical treatments of population genetics (Fisher 1930, Wright 1931).

Across these fields, mathematical models, laboratory experiments, and field studies have produced substantial mechanistic understanding of natural evolution. However, the direct study of emergent behavior in natural ecosystems is constrained by measurement, intervention, and ethical limitations. As ecosystems increase in spatial extent and interaction complexity, they become increasingly difficult to instrument and control; when such control is feasible, large-scale interventions can damage or displace wild populations.

To examine complex behavior induced by competitive and environmental pressures under controlled conditions, we study the open-ended evolution of neural network policies in large-scale ecological simulations. Agents in this setting are not assigned an externally specified objective or reward signal; rather, they collect resources required for survival and reproduction according to the transition dynamics of the environment. Policy variation occurs only through mutation when a parent transfers its policy parameters to an offspring. As computational resources available to researchers continue

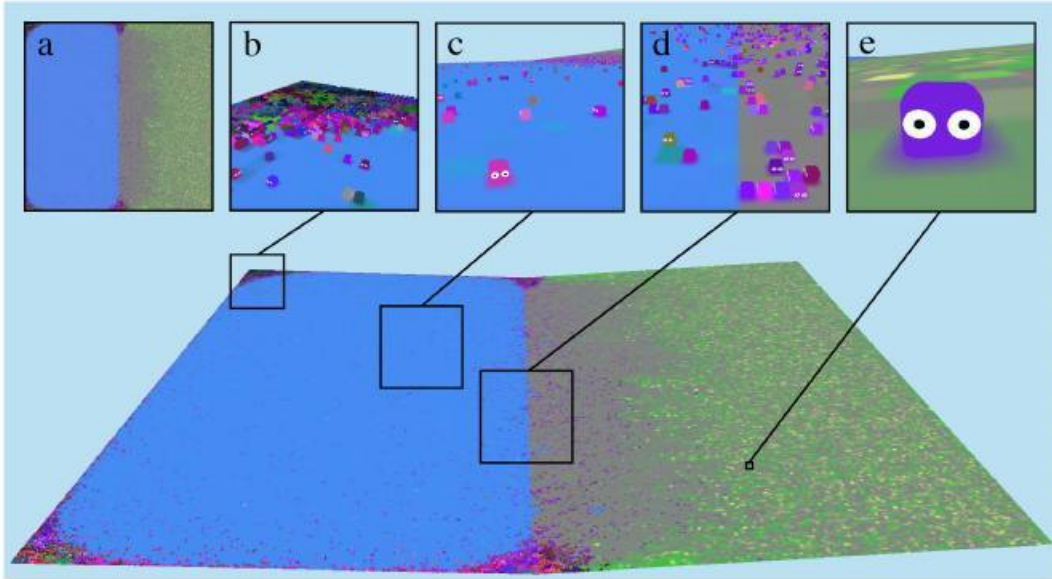


Figure 1: 3D visualization of a  $512 \times 512$  map in the simulation environment. Insert (a) shows a top-down view of the full map, (b) shows a cluster of agents in the corner region, (c) shows agents occupying water cells, (d) shows agents on the shoreline, (e) shows an agent near a tree in a sparse forest, and (f) shows agents moving through a dense forest.

to increase, our aim is to evaluate open-ended evolutionary dynamics in populations of simple digital organisms at previously inaccessible scales, thereby providing experimental tools for isolating the effects of environmental scale and population size on behavioral emergence.

We instantiate this study in a new JAX-based simulation environment that supports rapid evaluation of large grid worlds. The largest experiments contain more than 60,000 individual agents and more than 1,000,000 spatial grid cells. Agents must navigate the environment to acquire food and other resources required for survival, and they reproduce after collecting sufficient resources to instantiate a mutated copy of themselves. Figure 1 provides a 3D visualization of the environment.

Our objective is not only to use simulation to analyze ecological phenomena, but also to evaluate ecology as a mechanism for generating machine intelligence. Recent work in AI has shown that scale can do more than improve model performance: it can qualitatively alter model capabilities (Wei et al., 2022a). Despite this observation, many embodied-learning studies remain restricted to comparatively small environments and low-population regimes. We therefore ask what capabilities emerge when the scaled object is not a single model but an entire ecosystem. Analogously to reasoning behaviors that appear in individual models only beyond particular scale thresholds (Wei et al., 2022b), we seek to determine which strategies evolve in populations only when environmental size and complexity become sufficiently rich.

Across extensive experiments, we identify multiple cases in which sensor configuration and large environmental scale strongly affect simulated behavioral dynamics and produce long-term ecological consequences. For example, agents equipped with simple compass sensors adapt to perform long-range resource-gathering expeditions, but this behavior emerges less reliably in small environments. We also find that agents with vision sensors evolve more effective foraging and attacking behavior than agents without vision, with the same effects appearing more consistently at larger scales. These experiments indicate that the scale and complexity made practical by modern hardware accelerators provide a technically useful regime for studying how evolution in ecological environments can generate rich behavioral repertoires.

## 2 RELATED WORK

The emergence of life has motivated computational models since the beginning of the field. Conway’s Game of Life (Games 1970) specifies population dynamics through simple local transition rules. Early artificial-life systems such as Tierra (Ray 1992) and Avida (Brown 1992) represent life as self-replicating programs whose lineages evolve through mutation. Similarly, Reynolds (1987) demonstrated that local motion rules can generate macroscopic flocking and schooling behavior.

These principles have been used to evolve virtual-agent locomotion morphology (Sims 1994a,b; Ha 2018; Gupta et al. 2021; Auerbach et al. 2014; Auerbach & Bongard 2012; Jeon et al. 2025; Auerbach & Bongard 2014; Szerlip & Stanley 2013) and sensing systems (Pratt et al. 2022; Mugan & MacIver 2020; Tiwary et al. 2025), while neuroevolution methods have directly optimized neural network architectures (Stanley & Miikkulainen 2002, 2004; Stanley et al. 2009). In contrast, our experiments do not optimize agents for improved adaptation with respect to an explicit reward or objective signal. Instead, we analyze which behaviors arise from ecological pressure itself and how those behaviors depend on environmental scale and complexity.

Evolutionary principles have also informed the internal dynamics of optimization algorithms. Genetic algorithms (Holland 1992; Jong 1975) and evolutionary strategies (Wierstra et al. 2014; Hansen & Ostermeier 2001; Rechenberg 1973; Schwefel 1995; Conti et al. 2018) instantiate selection and variation as mechanisms for search. Related methods use evolutionary operators to encourage novelty among candidate solutions (Lehman & Stanley 2011; Cully et al. 2015). Other work applies such methods to optimize neural networks for tasks such as Atari games and Mujoco locomotion tasks (Salimans et al. 2017; Such et al. 2017; Mania et al. 2018). Although our system uses evolutionary mechanisms to update agent policies, our contribution is not a new optimizer and does not use evolution to solve a fixed objective; rather, it uses mutation, reproduction, and selection as the population-level dynamics whose emergent consequences are measured.

A further line of work uses biological inspiration to construct organisms situated within simulated environments. Early examples include Bedau (1992), Yeager et al. (1994), Channon & Dampier (1998), and Tisue et al. (2004). Recent systems use cellular automata to produce organisms with emergent morphologies and behaviors (Mordvintsev et al. 2020; Robinson 2021; Heinemann 2021; Randazzo & Mordvintsev 2023; Caussan 2025). Mordatch & Abbeel (2018) and Park et al. (2023) use simulated environments to study the emergence of language and communication among agents. Autocurricula (Leibo et al. 2019), in which co-adapting populations generate an increasingly challenging learning landscape for one another, have also been studied in environments with smaller populations than ours (Leibo et al. 2018; Team et al. 2021). In addition, Wang et al. (2019) and Aki et al. (2024) invert the emphasis by evolving environments that are optimized for agent learning. Although these systems evolve the environment rather than the agent, they motivate our focus on environmental complexity as a driver of sophisticated behavior.

The closest prior work simulates populations of agents interacting in complex environments. Baker et al. (2019) analyze a Hide-and-Seek task in which a small number of hider agents learn to use environmental tools to evade seeker agents, with both groups trained by reinforcement learning. Other systems similarly train populations for specified tasks, including playing video games (Jaderberg et al. 2019), manipulating objects (Petrenko et al. 2021), and capturing agents (Zheng et al. 2017; Yamada et al. 2020). However, these studies typically analyze emergence under fixed tasks and explicit objectives. The Neural MMO framework (Suarez et al. 2019, 2023) is more open-ended in that it places agents in a massively multi-agent game and optimizes primarily for survival rather than for a narrow task. However, its population is capped at a fixed size, with experiments considering up to 128 concurrent agents and without generational inheritance, which limits analysis of endogenous population dynamics. Hamon et al. (2023) is closer to our setting because it includes variable population sizes, genetically inspired evolution, and generational timelines, but its environment is substantially smaller, less feature-rich, and does not directly isolate the role of scale. Similarly, Lu et al. (2024) simulates agents that evolve to collect resources. However, that system also operates at a much smaller maximum scale of 256 agents and emphasizes tool use through programmable robots. Collectively, these studies establish important components of emergent ecological simulation, but a more complete analysis requires populations that can expand and contract endogenously, evolve without predefined objectives, and interact in environments sufficiently rich and large to sup-

port diverse collective strategies. Our work combines these properties, enabling measurement of ecological phenomena that were inaccessible in prior systems.

Finally, we draw on recent work emphasizing environment throughput and scale through optimized libraries that increase simulation speed and the number of agents that can interact in game-like settings (Koyamada et al., 2023; Matthews et al., 2024; Lechner et al., 2023). We build on this computational-efficiency perspective by using JAX-based acceleration techniques to obtain large-scale, high-throughput ecological simulation.

### 3 METHOD

#### 3.1 ECOLOGICAL GAMES

We formalize the interaction between an agent population and its environment as a multiplayer dynamical system termed an **Ecological Game (EG)**. As in Markov decision processes (MDPs) for reinforcement learning and stochastic games (SGs) for multi-agent reinforcement learning (see Sutton et al., (1998) and Albrecht et al. (2024)), an EG specifies Markovian transition dynamics. Its distinguishing purpose is to represent variable-size populations in an objective-free setting. Like an SG, an EG contains a set of agents  $I$ , a state space  $S$ , an initial state distribution  $\rho_0$ , an action space  $A$ , and a transition function  $\mathcal{T}$  mapping a state  $s_t$  and each player action  $a_{t,i}$  to a distribution over next states  $s_{t+1}$ . Unlike an SG, an EG omits the reward function and introduces a population function  $\Gamma(s_t)$  that records which agents in  $I$  are alive and the parent identity of each agent, allowing birth and death events to be inferred from state. As MDPs and SGs admit partially observable variants, POMDPs and POSGs, an EG may also be partially observable; we refer to this case as a POEG. A POEG differs from an EG by replacing full-state access with an observation function  $O(s_t, i)$  that maps a state and player to that player’s observation. The environment used in this work is a POEG whose observation function depends on the sensor configuration specified below.

#### 3.2 ENVIRONMENT

The experimental POEG is a large grid world containing resources that agents must acquire to satisfy survival and reproduction constraints. The simulator maintains a factored state  $s_t = \{s_t^m, s_t^a\} \in S = S^m \times S^a$ , where  $s_t^m$  contains two-dimensional map variables such as terrain and resource fields, and  $s_t^a$  contains agent-indexed variables such as position, health, and internal resources.

**Terrain:** Each map includes a rock layer that defines the elevation of every grid cell. Water is initialized on cells whose rock value lies below a fixed sea level. During an episode, water is updated by a discrete downhill-flow simulation whose potential field is the rock-layer height. The experiments use the six global terrain configurations shown in Figure 2: **Ocean**, **Beach**, **Island**, **Lake**, **Isthmus**, and **Channel**. We evaluate grid sizes of  $64 \times 64$ ,  $128 \times 128$ ,  $256 \times 256$ ,  $512 \times 512$ , and  $1024 \times 1024$ . When the grid size changes, the initial population is scaled proportionally to total map area; after initialization, population size evolves endogenously according to environmental dynamics.

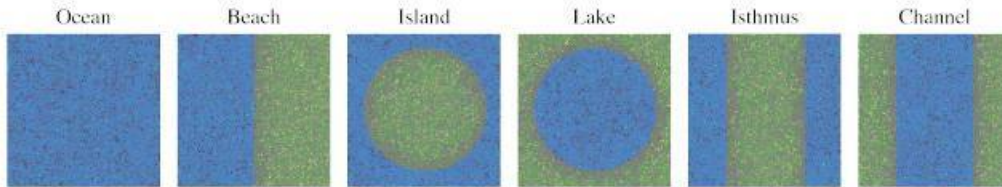


Figure 2: Overhead renderings of the terrain configurations used in the experiments at  $256 \times 256$  resolution. Small dark marks correspond to individual agents, while green and yellow indicate local concentrations of energy and biomass. Each image was captured 5000 steps into an episode, after most water-based biomass had been consumed and before agents had evolved extended survival on land.

**Resources and Agent Health:** Agents must acquire **water**, **energy**, and **biomass** to remain alive and to reproduce. At initialization, biomass and energy are randomly distributed over the environment, after which individual agents consume and redistribute these resources through their actions.

Every agent action incurs a small water and energy cost. Spent water is returned to the terrain and subsequently flows downhill toward the nearest body of water, whereas spent energy is removed from the system. Biomass controls reproduction: each agent requires a fixed biomass allocation to exist and therefore can reproduce only after accumulating excess biomass that can be transferred to offspring. Total biomass is conserved over the simulation, as water is, but biomass neither flows downhill nor moves autonomously. Energy is non-conserved and slowly accumulates on free-standing biomass. These resource rules are designed to create a multi-resource control problem that is more structured than single-food collection. For ablations on resource rules, see Appendix F.2

Agents also maintain health points (HP), which determine viability. If an agent overspends a critical resource, its HP decreases rapidly; conversely, agents can expend energy and water to recover HP gradually over time. Agents may attack other agents, thereby reducing target HP. They also receive a small age-dependent damage term that increases with age, so agents eventually die of senescence if no other cause intervenes. When an agent’s HP reaches zero, the resources it carries are deposited into the grid cell it occupies. Consequently, an agent that attacks and kills another agent may subsequently consume the released resources, implementing a minimal form of predation. For ablations on HP hyperparameters, see Appendix F.8

**Sensors:** Agents observe the POEG through four sensor types. The **internal** sensor reports the agent’s age, health, and carried resources; the **external** sensor reports resources located in the agent’s current grid cell; the **compass**, motivated by magnetoreception in birds and other animals, encodes the agent’s global orientation as a 4-way one-hot vector; and **vision** provides a  $7 \times 7$  top-down rendered patch centered on the agent, with three color channels and a fourth channel encoding local elevation differences. For ablations on vision configurations, see Appendix F.6

**Actions:** The action space is finite and consists of five action categories. **Rest** performs no environmental operation and has the lowest energy cost. The three **Move** actions rotate the agent clockwise, rotate it counterclockwise, or advance it one grid cell. Grid occupancy is exclusive: if an agent attempts to enter an occupied cell, or if multiple agents attempt to enter the same cell, the corresponding move action is canceled. The **Attack** action kills all agents in the  $3 \times 3$  square directly in front of the acting agent. For ablations on attack strength, see Appendix F.7 The three **Eat** actions correspond to the three resource types. If the agent occupies a cell containing the selected resource, the corresponding action transfers a fixed amount of that resource from the environment into the agent’s stomach, subject to a fixed maximum capacity.

Finally, agents have a **Reproduce** action implementing single-parent reproduction. This action succeeds only when the parent has accumulated sufficient resources to allocate to an offspring. The new agent is instantiated directly behind the parent. Its policy is a mutated copy of the parent’s weights, and it immediately inherits a fixed quantity of the parent’s resources.

In the experiments, we selectively disable sensors or the attack action to isolate the ecological consequences of different agent capabilities. We use three sensor configurations: internal+external resources only (**R**), internal+external+compass (**RC**), and internal+external+compass+vision (**RCV**). The suffix **+A** denotes that attacking is enabled; for example, (**RC+A**) denotes an agent with resource sensors, a compass, and the attack action.

### 3.3 POLICIES

Agent policies are implemented as small memoryless MLPs. We exclude memory to reduce the computational model and parameter count. For each sensor, raw values are flattened, normalized to  $[-1, 1]$ , and passed through a separate linear layer with output dimension 64. The resulting embeddings are summed and provided to a 2-layer MLP with hidden dimension 64 and ReLU nonlinearities. Because the action space is discrete, the final layer is linear with output dimension equal to the number of actions. Actions are sampled from a softmax distribution whose temperature evolves together with the network weights. Network weights are stored as half-precision bfloat16 values to support larger population sizes and shorter wall-clock runtimes. Depending on the enabled sensors, each agent network contains approximately 10k-25k parameters. For ablations on network size and architecture, see Appendix F.5

For the initial population in every run, weights are sampled with Kaiming initialization (He et al., 2015), and bias vectors are initialized to zero. When an agent is born during an episode, its policy

is produced by copying the parent’s weights and adding Gaussian noise with standard deviation  $3 \times 10^{-2}$  to every mutable weight value. This mutation rule is similar to mechanisms used in prior neuroevolution work (Such et al., 2017). Although more complex reproductive operators such as crossover are possible, we use this simple mechanism to keep reproduction minimal and to isolate the effects of environmental scale. Empirically, the use of low-precision floating-point values required a larger mutation rate. We report a mutation-rate ablation in Appendix F.4.

Each agent also has a 3-channel trait that controls its rendered color on the map. This color trait mutates under the same mechanism as the network weights and determines the agent’s appearance to other agents with the vision sensor. In principle, the trait can support visual discrimination among different agent types.

### 3.4 COMPUTATION AND REPRODUCIBILITY

All experiments are implemented in JAX (Bradbury et al., 2018) and executed on Nvidia H100 and A40 GPUs using a state-of-the-art simulation environment. Unless otherwise specified, each experiment runs for 2M environment steps or until population extinction. Per-seed wall-clock time ranges from less than one hour for  $64 \times 64$  grids to approximately 10 hours for  $1024 \times 1024$  grids on a single H100 GPU. Appendix E provides additional simulator-architecture details, scaling measurements, and reproducibility considerations.

## 4 EXPERIMENTS

The experiments isolate how sensor modality, environmental scale, and population size affect the behavioral dynamics that emerge from the ecological simulator.

### 4.1 LONG-DISTANCE RESOURCE GATHERING

The first experiment evaluates whether agents can evolve policies that traverse long inland distances to locate resources. In environments with a clear directional structure, agents equipped with a compass (RC) consistently develop more effective long-range resource-gathering behavior than baseline agents without a compass (R). The same behavior also emerges more reliably and with greater functional impact at larger environmental scales.

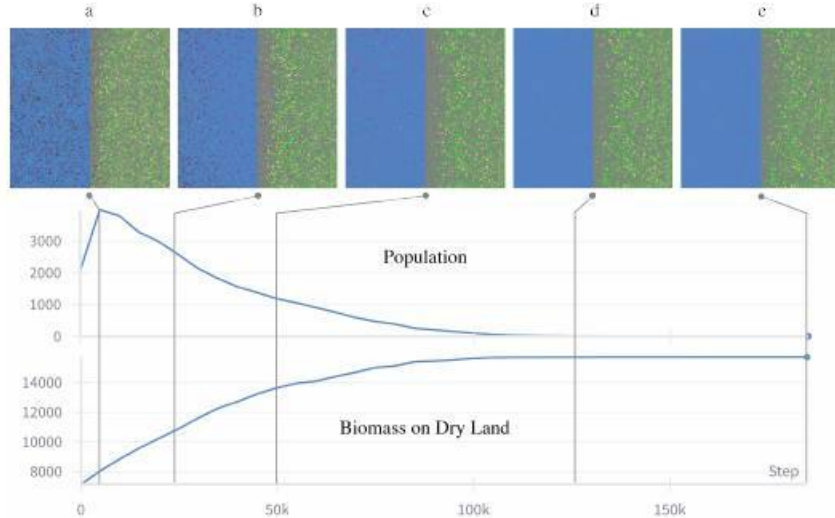


Figure 3: Representative trial with (R) agents in a  $256 \times 256$  Beach environment. The top panel reports population size over time, and the bottom panel reports free biomass located on dry land.

We first analyze a representative (R) run, shown in Figure 3. After 5000 time steps (a), all initially land-based agents have died because they lack access to water, while water-based agents have ex-

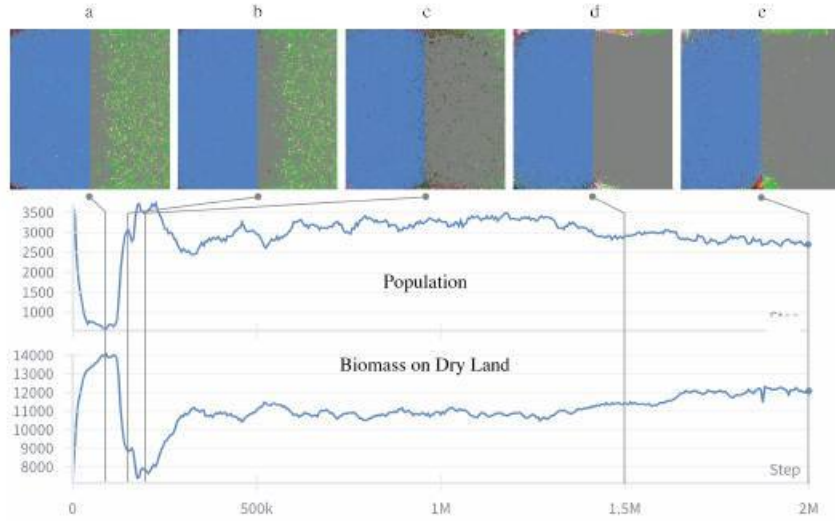


Figure 4: Representative trial with **(RC)** agents in a  $256 \times 256$  **Beach** environment. The top panel reports population size over time, and the bottom panel reports free biomass located on dry land.

panded rapidly due to abundant initial resources. At this early stage, the population has not yet evolved policies that efficiently search for resources. Following this early population peak, some water-based agents occasionally move onto land (b). These agents often fail to navigate back to the ocean and die, depositing their carried biomass on land. By 50,000 time steps (c), this cumulative process has nearly doubled the quantity of uneaten land biomass. Because biomass has been transferred from water to land, the water region can no longer support the initial large population, causing the population to contract. By 125,000 steps (d), almost all biomass has been transferred to land and the population is near extinction. The final agent dies at approximately 185,000 steps (e).

This trajectory illustrates a population-level consequence of the agents’ inability to navigate back to water. Individual deaths alter the spatial resource distribution, and those changes feed back into the viability of the entire population. At the same time, the resulting land-based biomass deposits create a selective opportunity for any agent that can move onto land, exploit the accumulated biomass, and return safely to water.

Figure 4 shows the corresponding behavior when agents are given an internal compass **(RC)**. Initially, the run follows a similar pattern (a): population size decreases and free biomass on land increases rapidly. In this configuration, however, some agents use the compass to drift toward the relative safety of map corners. At approximately 150,000 steps (b), a subset of agents begins to “mine” the land biomass by using compass information to navigate repeatedly between land and water. By 200,000 steps (c), these miners have cleared the beach. The population subsequently exhibits minor structural changes, including clumping near map boundaries (d), but remains largely stable through the full 2,000,000 time-step horizon (e).

Table 1: Mining and extinction events in the **Beach** world for compass-equipped populations **(RC)** and populations without a compass **(R)** across five world sizes and initial population sizes. denotes the number of seeds with a mining event. denotes the number of seeds in which the population went extinct before 2M steps. Table 3 in Appendix A reports seed-level data.

|             | 1024 × 1024 |     | 512 × 512 |     | 256 × 256 |     | 128 × 128 |     | 64 × 64 |     |
|-------------|-------------|-----|-----------|-----|-----------|-----|-----------|-----|---------|-----|
|             | 32768       |     | 8192      |     | 2048      |     | 512       |     | 128     |     |
|             |             |     |           |     |           |     |           |     |         |     |
| <b>(RC)</b> | 4/4         | 0/4 | 4/4       | 0/4 | 4/4       | 0/4 | 3/4       | 1/4 | 2/4     | 3/4 |
| <b>(R)</b>  | 1/4         | 0/4 | 0/4       | 4/4 | 0/4       | 4/4 | 0/4       | 4/4 | 0/4     | 4/4 |

We quantify these observations by running the full **Beach** experiment suite at grid sizes  $1024 \times 1024$ ,  $512 \times 512$ ,  $256 \times 256$ ,  $128 \times 128$ , and  $64 \times 64$ , with initial populations of 32768, 8192, 2048, 512, and 128 agents, respectively. Each configuration is evaluated with four random seeds. We record whether mining behavior emerges, operationalized as a decrease of at least 10% in dry-land free biomass relative to its historical maximum. We also record the mining-event onset time, defined by the local maximum in land biomass, and the corresponding percentage decrease in biomass. Table 1 summarizes mining and extinction events; Table 3 in Appendix A reports seed-level details. Mining emerges more consistently for (RC) agents than for (R) agents. For (RC) agents, mining is also more consistent in larger environments and appears in every seed from the  $256 \times 256$  scale upward. At smaller scales, populations are less stable and have a higher probability of early extinction.

To analyze the temporal structure of these effects, we run additional (R) and (RC) seeds at  $512 \times 512$  world size (32 seeds) and  $128 \times 128$  world size (64 seeds). We use a Kaplan-Meier estimator to estimate the probability of extinction and mining as functions of simulator time. Extinction is rarer and occurs later in larger worlds, whereas mining is more frequent and begins earlier in smaller grids. We also fit a Cox proportional hazards model comparing mining-discovery time and extinction time across world size and compass usage; this analysis similarly indicates that larger world size substantially lowers extinction rate and enables much faster mining discovery than smaller worlds. Enabling the compass produces a comparable effect. Full details are provided in Appendix B.

Table 2: Mining and extinction events in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains for compass-equipped populations (RC) and populations without a compass (R) at  $512 \times 512$  resolution.  denotes the number of seeds with a mining event.  denotes the number of seeds in which the population went extinct before 2M steps. Table 4 in Appendix A reports seed-level data.

|      | Island                                                                             |                                                                                    | Lake                                                                               |                                                                                    | Isthmus                                                                            |                                                                                    | Channel                                                                            |                                                                                      |
|------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|      |  |  |  |  |  |  |  |  |
| (RC) | 4/4                                                                                | 0/4                                                                                | 2/4                                                                                | 2/4                                                                                | 4/4                                                                                | 0/4                                                                                | 3/4                                                                                | 1/4                                                                                  |
| (R)  | 3/4                                                                                | 1/4                                                                                | 2/4                                                                                | 4/4                                                                                | 1/4                                                                                | 3/4                                                                                | 0/4                                                                                | 4/4                                                                                  |

We then repeat the analysis in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains at  $512 \times 512$  resolution using both (R) and (RC) agents to evaluate the dependence of mining on global terrain geometry. Table 2 reports mining and extinction events, and Table 4 in Appendix A provides run-level details. As in the **Beach** environment, compass access yields earlier and more consistent mining and produces more stable populations with lower extinction rates. (RC) agents adapt more readily to the island and isthmus terrains than to the lake and channel terrains, where extinction occurs in some seeds. We hypothesize that this difference arises because moving consistently east or west returns an agent to water in the island and isthmus terrains, whereas the channel and lake terrains do not provide an equally simple directional rule. The (R) configuration does not reliably adapt to any of these terrains, although it performs least poorly on the island. Figure 12 in Appendix D visualizes mining in the island terrain.

## 4.2 VISUAL FORAGING AND PREDATION

We next evaluate whether agents in the **Ocean** environment can use vision to locate free resources and prey on other agents. We compare (RCV+A) agents, which have resource sensors, a compass, a  $7 \times 7$  visual sensor, and attacking enabled, against (RC+A) agents, which have only resource sensors, a compass, and attacking enabled.

Figure 5 reports population size, biomass utilization, move-action frequency, and eat-action frequency for  $512 \times 512$  and  $128 \times 128$  grids. Additional grid sizes are shown in Figure 9 in Appendix C and summary statistics are reported in Table 5 in Appendix A. Biomass utilization is defined as the percentage of total environmental biomass currently carried by agents. In runs with vision sensors (red curves), agents spend less time moving and more time eating. These runs also sustain larger populations and achieve higher biomass utilization. The resulting dynamics indicate that visual sensing is being used to produce more efficient foraging behavior. Across environmental scales,

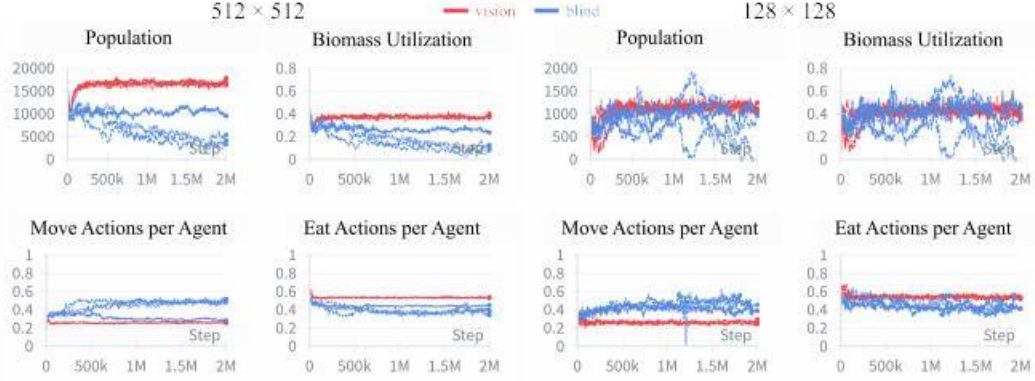


Figure 5: Population size, biomass utilization, move actions per agent, and eat actions per agent in  $512 \times 512$  and  $128 \times 128$  **Ocean** worlds. Biomass utilization, move actions per agent, and eat actions per agent are averaged over agents at each time step. Each panel contains four seeds with visual agents (RCV + A) (red) and four seeds with blind agents (RC + A) (blue). Vision yields larger and more stable populations, higher biomass utilization, less movement, and more frequent effective eating. At larger scales, these effects exhibit lower cross-seed variance, indicating that vision provides an increasing advantage over blindness for adaptive foraging. See Figure 9 in Appendix C for the corresponding  $1024 \times 1024$  and  $256 \times 256$  comparison.

between-run variance decreases substantially in larger environments. Runs with vision also exhibit markedly more stable temporal dynamics, with lower variation in the reported metrics over time.

Figure 6 reports attacks per agent and homicides per attack for the same runs. Agents with vision attack relatively infrequently but maintain a kill rate of approximately 75%. In large grids, blind agents sometimes also exhibit high kill rates, but these trajectories are substantially less stable. Several confounding factors prevent a definitive causal interpretation of this observation. Visual inspection (Figure 13 in Appendix D) indicates that high kill rates for blind agents sometimes coincide with dense spatial clusters, whereas such clusters are rarely observed in visual-agent runs. This pattern may indicate that visual agents also adapt to avoid one another. In smaller environments, the difference between visual and blind agents is less pronounced, but visual-agent populations remain substantially more stable.

### 4.3 ABLATIONS

Appendix F reports ablations over initial population size (F.1), resource rules (F.2), mutation rate (F.4), agent network configuration (F.5), vision range (F.6), attack strength (F.7), and agent HP (F.8), thereby measuring the sensitivity of emergent behavior to simulator configuration. Appendix E.3 further tests whether the dynamics observed in larger environments can be explained solely by self-averaging.

## 5 CONCLUSION

This study demonstrates that open-ended ecological environments constitute a technically useful paradigm for analyzing the emergence of intelligence. Across experiments that vary world scale and environmental complexity, we find that both spatial scale and sensory richness materially affect the evolution of sensing and decision-making policies. Taken together, the results suggest that ecological and evolutionary pressures, when embedded in sufficiently rich environments, may provide a natural mechanism for producing increasingly capable artificial agents. The experiments also show that nontrivial behavioral repertoires can be evolved in large populations under a feasible hardware budget without specifying explicit objectives. We expect these findings to support further investigation of ecology as both a generative mechanism for artificial intelligence and an analysis framework for understanding it.

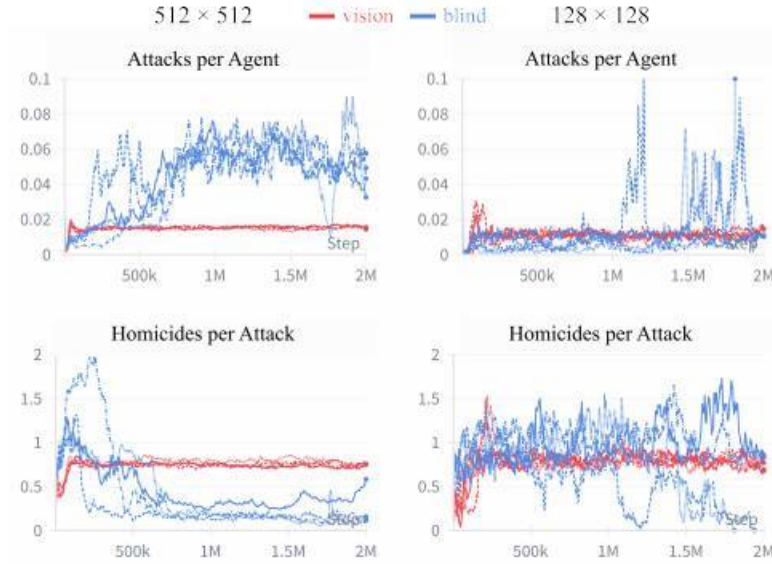


Figure 6: Attacks per agent and homicides per attack, averaged over agents, in  $512 \times 512$  and  $128 \times 128$  **Ocean** worlds. Each panel contains four seeds with visual agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). With vision, agents attack less frequently but with substantially higher precision, consistent with specialized predation behavior. This effect is stronger at larger scales. See Figure 10 in Appendix C for the corresponding  $1024 \times 1024$  and  $256 \times 256$  comparison.

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## A ADDITIONAL DATA

Table 3: Mining events in the **Beach** world for compass-equipped agents (**RC**) and agents without a compass (**R**) across five spatial scales.  indicates whether a run produced a mining event.  reports the percentage decrease in free dry-land biomass during the mining event, and  reports the approximate timestep at which free dry-land biomass reached the local maximum defining event onset. For runs with multiple mining events, the first event is reported.  reports the extinction timestep, or n/a if the run persisted for 2M steps.

|                 | 1024x1024                                                                         |                                                                                   |                                                                                   |                                                                                   | 512x512                                                                           |                                                                                   |                                                                                   |                                                                                   | 256x256                                                                           |                                                                                   |                                                                                   |                                                                                   | 128x128                                                                           |                                                                                   |                                                                                   |                                                                                   | 64x64 |  |  |  |
|-----------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------|--|--|--|
|                 | 32768                                                                             |                                                                                   |                                                                                   |                                                                                   | 8192                                                                              |                                                                                   |                                                                                   |                                                                                   | 2048                                                                              |                                                                                   |                                                                                   |                                                                                   | 512                                                                               |                                                                                   |                                                                                   |                                                                                   | 128   |  |  |  |
| Compass On (RC) |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |       |  |  |  |
| Seed            |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |
| 0               | ✓                                                                                 | 35k                                                                               | 11.1                                                                              | n/a                                                                               | ✓                                                                                 | 105k                                                                              | 33.                                                                               | n/a                                                                               | ✓                                                                                 | 145k                                                                              | 49.3                                                                              | n/a                                                                               | ✓                                                                                 | 105k                                                                              | 72.5                                                                              | n/a                                                                               | x     |  |  |  |
| 1               | ✓                                                                                 | 90k                                                                               | 16.2                                                                              | n/a                                                                               | ✓                                                                                 | 50k                                                                               | 17.2                                                                              | n/a                                                                               | ✓                                                                                 | 50k                                                                               | 46.6                                                                              | n/a                                                                               | x                                                                                 |                                                                                   | 140                                                                               | ✓                                                                                 | 70k   |  |  |  |
| 2               | ✓                                                                                 | 40k                                                                               | 15.                                                                               | n/a                                                                               | ✓                                                                                 | 30k                                                                               | 14.3                                                                              | n/a                                                                               | ✓                                                                                 | 55k                                                                               | 34.7                                                                              | n/a                                                                               | ✓                                                                                 | 50k                                                                               | 50.7                                                                              | n/a                                                                               | ✓     |  |  |  |
| 3               | ✓                                                                                 | 60k                                                                               | 15.                                                                               | n/a                                                                               | ✓                                                                                 | 45k                                                                               | 15.6                                                                              | n/a                                                                               | ✓                                                                                 | 90k                                                                               | 47.4                                                                              | n/a                                                                               | ✓                                                                                 | 20k                                                                               | 52.                                                                               | n/a                                                                               | x     |  |  |  |
| Avg             | 4/4                                                                               | 34k                                                                               | 14.3                                                                              | 0/4                                                                               | 4/4                                                                               | 58k                                                                               | 20.                                                                               | 0/4                                                                               | 4/4                                                                               | 85k                                                                               | 44.5                                                                              | 0/4                                                                               | 3/4                                                                               | 58k                                                                               | 58.4                                                                              | 1/4                                                                               | 2/4   |  |  |  |
| Compass Off (R) |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |       |  |  |  |
| Seed            |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |
| 0               | x                                                                                 |                                                                                   |                                                                                   | n/a                                                                               | x                                                                                 |                                                                                   |                                                                                   | 945k                                                                              | x                                                                                 |                                                                                   |                                                                                   | 185k                                                                              | x                                                                                 |                                                                                   |                                                                                   | 65k                                                                               | x     |  |  |  |
| 1               | ✓                                                                                 | 275k                                                                              | 12.4                                                                              | n/a                                                                               | x                                                                                 |                                                                                   |                                                                                   | 1085k                                                                             | x                                                                                 |                                                                                   |                                                                                   | 110k                                                                              | x                                                                                 |                                                                                   |                                                                                   | 55k                                                                               | x     |  |  |  |
| 2               | x                                                                                 |                                                                                   |                                                                                   | n/a                                                                               | x                                                                                 |                                                                                   |                                                                                   | 1070k                                                                             | x                                                                                 |                                                                                   |                                                                                   | 165k                                                                              | x                                                                                 |                                                                                   |                                                                                   | 55k                                                                               | x     |  |  |  |
| 3               | x                                                                                 |                                                                                   |                                                                                   | n/a                                                                               | x                                                                                 |                                                                                   |                                                                                   | 280k                                                                              | x                                                                                 |                                                                                   |                                                                                   | 125k                                                                              | x                                                                                 |                                                                                   |                                                                                   | 90k                                                                               | x     |  |  |  |
| Avg             | 1/4                                                                               | 275k                                                                              | 12.4                                                                              | 0/4                                                                               | 0/4                                                                               |                                                                                   |                                                                                   | n/a                                                                               | 0/4                                                                               |                                                                                   |                                                                                   | 4/4                                                                               | 0/4                                                                               |                                                                                   |                                                                                   | 4/4                                                                               | 0/4   |  |  |  |

Table 4: Mining events in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains for **(RC)** agents relative to **(R)** agents. All runs use a  $512 \times 512$  grid and an initial population of 8192 agents.

|                 | Island |      |      |      | Lake |     |      |       | Isthmus |     |      |      | Channel |     |      |      |
|-----------------|--------|------|------|------|------|-----|------|-------|---------|-----|------|------|---------|-----|------|------|
| Compass On (RC) |        |      |      |      |      |     |      |       |         |     |      |      |         |     |      |      |
| Seed            |        |      |      |      |      |     |      |       |         |     |      |      |         |     |      |      |
| 0               | ✓      | 40k  | 94.3 | n/a  | x    |     |      | 45k   | ✓       | 45k | 33.  | n/a  | ✓       | 50k | 27.4 | n/a  |
| 1               | ✓      | 30k  | 94.3 | n/a  | ✓    | 30k | 22.5 | n/a   | ✓       | 55k | 37.3 | n/a  | ✓       | 80k | 30.8 | n/a  |
| 2               | ✓      | 30k  | 94.8 | n/a  | x    |     |      | 45k   | ✓       | 40k | 28.1 | n/a  | ✓       | 50k | 41.  | n/a  |
| 3               | ✓      | 70k  | 95.1 | n/a  | ✓    | 35k | 25.3 | n/a   | ✓       | 30k | 41.5 | n/a  | x       |     |      | 80k  |
| Avg             | 4/4    | 43k  | 94.6 | 0/4  | 2/4  | 33k | 23.9 | 2/4   | 4/4     | 43k | 35.  | 0/4  | 3/4     | 60k | 33.1 | 1/4  |
| Compass Off (R) |        |      |      |      |      |     |      |       |         |     |      |      |         |     |      |      |
| Seed            |        |      |      |      |      |     |      |       |         |     |      |      |         |     |      |      |
| 0               | x      |      |      | 185k | x    |     |      | 125k  | x       |     |      | 145k | x       |     |      | 100k |
| 1               | ✓      | 75k  | 30.9 | n/a  | x    |     |      | 130k  | x       |     |      | 260k | x       |     |      | 105k |
| 2               | ✓      | 85k  | 47.  | n/a  | ✓    | 75k | 23.4 | 1400k | ✓       | 95k | 18.8 | n/a  | x       |     |      | 105k |
| 3               | ✓      | 115k | 39.5 | n/a  | ✓    | 70k | 24.3 | 1250k | x       |     |      | 140k | x       |     |      | 100k |
| Avg             | 3/4    | 92k  | 39.1 | 1/4  | 2/4  | 73k | 23.9 | 4/4   | 1/4     | 95k | 18.8 | 3/4  | 0/4     |     |      | 4/4  |

Table 5: Population and foraging measurements for visual and blind agents in the **Ocean** world across scales. denotes population size. denotes biomass utilization, the percentage of total system biomass currently inside agent stomachs. denotes move actions per agent, defined as the fraction of time an agent selected a move action and averaged across agents. similarly denotes the proportion of agent actions that were eat actions. Reported , , , and values are seed-level averages over all steps.

|                | 1024x1024<br>32768                                                                  |                                                                                     |                                                                                     |                                                                                     | 512x512<br>8192                                                                     |                                                                                     |                                                                                     |                                                                                     | 256x256<br>2048                                                                     |                                                                                     |                                                                                      |                                                                                       | 128x128<br>512                                                                        |                                                                                       |                                                                                       |                                                                                       |
|----------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Vision (RCV+A) |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |
| Seed           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 0              | 58354                                                                               | .396                                                                                | .402                                                                                | .426                                                                                | 16413                                                                               | .369                                                                                | .257                                                                                | .532                                                                                | 4195                                                                                | .382                                                                                | .255                                                                                 | .536                                                                                  | 1126                                                                                  | .426                                                                                  | .261                                                                                  | .541                                                                                  |
| 1              | 65107                                                                               | .365                                                                                | .257                                                                                | .529                                                                                | 16200                                                                               | .364                                                                                | .256                                                                                | .533                                                                                | 4142                                                                                | .377                                                                                | .258                                                                                 | .536                                                                                  | 1103                                                                                  | .415                                                                                  | .262                                                                                  | .537                                                                                  |
| 2              | 64735                                                                               | .363                                                                                | .257                                                                                | .530                                                                                | 16560                                                                               | .372                                                                                | .257                                                                                | .533                                                                                | 4269                                                                                | .388                                                                                | .260                                                                                 | .536                                                                                  | 1041                                                                                  | .392                                                                                  | .254                                                                                  | .542                                                                                  |
| 3              | 64595                                                                               | .362                                                                                | .256                                                                                | .532                                                                                | 16206                                                                               | .365                                                                                | .255                                                                                | .531                                                                                | 4249                                                                                | .388                                                                                | .259                                                                                 | .533                                                                                  | 1100                                                                                  | .414                                                                                  | .257                                                                                  | .538                                                                                  |
| Avg            | 63198                                                                               | .371                                                                                | .293                                                                                | .504                                                                                | 16345                                                                               | .368                                                                                | .256                                                                                | .532                                                                                | 4213                                                                                | .384                                                                                | .258                                                                                 | .535                                                                                  | 1092                                                                                  | .412                                                                                  | .259                                                                                  | .539                                                                                  |
| Blind (RC+A)   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |
| Seed           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 0              | 22734                                                                               | .143                                                                                | .446                                                                                | .372                                                                                | 6549                                                                                | .173                                                                                | .442                                                                                | .400                                                                                | 2036                                                                                | .209                                                                                | .433                                                                                 | .461                                                                                  | 545                                                                                   | .219                                                                                  | .465                                                                                  | .427                                                                                  |
| 1              | 23670                                                                               | .151                                                                                | .426                                                                                | .388                                                                                | 6939                                                                                | .180                                                                                | .438                                                                                | .396                                                                                | 3107                                                                                | .321                                                                                | .390                                                                                 | .416                                                                                  | 1082                                                                                  | .444                                                                                  | .416                                                                                  | .460                                                                                  |
| 2              | 24745                                                                               | .159                                                                                | .358                                                                                | .426                                                                                | 5448                                                                                | .136                                                                                | .474                                                                                | .374                                                                                | 1587                                                                                | .163                                                                                | .440                                                                                 | .433                                                                                  | 979                                                                                   | .421                                                                                  | .404                                                                                  | .436                                                                                  |
| 3              | 20004                                                                               | .130                                                                                | .469                                                                                | .359                                                                                | 10354                                                                               | .265                                                                                | .311                                                                                | .439                                                                                | 3164                                                                                | .324                                                                                | .321                                                                                 | .450                                                                                  | 788                                                                                   | .317                                                                                  | .431                                                                                  | .477                                                                                  |
| Avg            | 22788                                                                               | .146                                                                                | .425                                                                                | .386                                                                                | 7323                                                                                | .189                                                                                | .416                                                                                | .402                                                                                | 2473                                                                                | .254                                                                                | .396                                                                                 | .440                                                                                  | 848                                                                                   | .351                                                                                  | .429                                                                                  | .450                                                                                  |

Table 6: Predation measurements for visual and blind agents in the **Ocean** world across scales.  denotes attacks per agent, defined as the fraction of time an agent selected the attack action and averaged across agents.  denotes homicides per attack, defined as the number of agents killed per attack action. Reported  and  values are averages over all steps for the seed. The 0.562  value for seed 3 in the  $128 \times 128$  (**RC+A**) run is an outlier: the seed went extinct before 2M steps, producing a spike in attacks per agent as population size decreased.

|                | 1024x1024                                                                           |                                                                                     | 512x512                                                                             |                                                                                     | 256x256                                                                             |                                                                                     | 128x128                                                                             |                                                                                       |
|----------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                | 32768                                                                               |                                                                                     | 8192                                                                                |                                                                                     | 2048                                                                                |                                                                                     | 512                                                                                 |                                                                                       |
| Vision (RCV+A) |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                       |
| Seed           |  |  |  |  |  |  |  |  |
| 0              | .011                                                                                | .776                                                                                | .015                                                                                | .731                                                                                | .013                                                                                | .784                                                                                | .010                                                                                | .788                                                                                  |
| 1              | .016                                                                                | .733                                                                                | .015                                                                                | .741                                                                                | .014                                                                                | .780                                                                                | .011                                                                                | .793                                                                                  |
| 2              | .016                                                                                | .707                                                                                | .015                                                                                | .737                                                                                | .014                                                                                | .750                                                                                | .012                                                                                | .771                                                                                  |
| 3              | .016                                                                                | .712                                                                                | .015                                                                                | .780                                                                                | .013                                                                                | .759                                                                                | .011                                                                                | .765                                                                                  |
| Avg            | .015                                                                                | .732                                                                                | .015                                                                                | .747                                                                                | .014                                                                                | .758                                                                                | .011                                                                                | .779                                                                                  |
| Blind (RC+A)   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                       |
| Seed           |  |  |  |  |  |  |  |  |
| 0              | .070                                                                                | .218                                                                                | .042                                                                                | .437                                                                                | .005                                                                                | 1.074                                                                               | .016                                                                                | .558                                                                                  |
| 1              | .054                                                                                | .300                                                                                | .044                                                                                | .383                                                                                | .016                                                                                | .871                                                                                | .006                                                                                | 1.032                                                                                 |
| 2              | .076                                                                                | .236                                                                                | .048                                                                                | .238                                                                                | .031                                                                                | .383                                                                                | .010                                                                                | .979                                                                                  |
| 3              | .059                                                                                | .203                                                                                | .044                                                                                | .456                                                                                | .033                                                                                | .446                                                                                | .562                                                                                | .841                                                                                  |
| Avg            | .065                                                                                | .239                                                                                | .045                                                                                | .378                                                                                | .021                                                                                | .693                                                                                | .148                                                                                | .853                                                                                  |

## B DEMOGRAPHY

We compare extinction and mining-emergence rates as functions of world size and agent access to global direction through the compass sensor. Extinction is defined as the population reaching zero individuals, and mining is defined as in the preceding experiments: a 10% decrease in free land biomass.

Runs are stratified by both world size and compass access. We simulate 64 runs each for  $128 \times 128$  without a compass and  $128 \times 128$  with a compass, and 32 runs each for  $512 \times 512$  without a compass and  $512 \times 512$  with a compass. For each of the four groups, we fit a non-parametric Kaplan–Meier estimator to the time-to-extinction distribution, treating runs that do not go extinct within 2M steps as right-censored. Figures 7 and 8 indicate that worlds tend to survive longer and produce mining earlier when they are large or when agents have compass information.

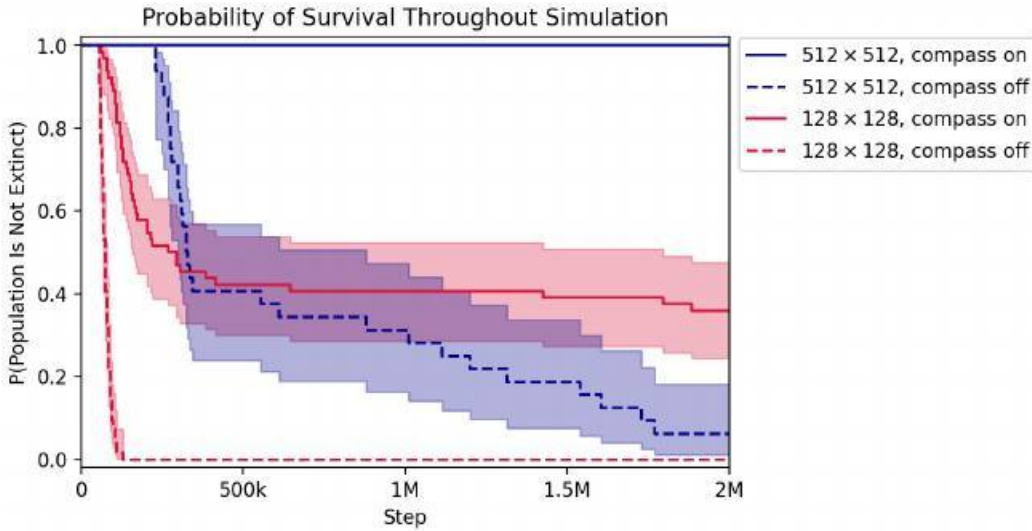


Figure 7: Estimated probability that the population has not gone extinct as a function of world size, compass access, and timestep. We fit a separate non-parametric Kaplan–Meier estimator for each run group (e.g.  $512 \times 512$ , compass on). Shaded regions denote 95% confidence intervals. Each  $512 \times 512$  group fits the estimator on 32 runs; each  $128 \times 128$  group fits on 64 runs.

We compare emergence and extinction timing with a Cox proportional hazards model, a semi-parametric survival model relating covariates to event hazard rates (Table 7). Runs that do not experience emergence or extinction by the end of the 2M-step horizon are treated as censored. For extinction, both larger world size and compass access substantially reduce the hazard; the extinction rate in  $128 \times 128$  worlds is approximately 25 times greater than the corresponding rate in  $512 \times 512$  worlds ( $HR=0.04$ ). Enabling the compass produces a similar-magnitude effect ( $HR=0.03$ ). For mining emergence, larger worlds generate mining much more rapidly ( $HR=6.07$ ), and compass access accelerates emergence dramatically ( $HR=34.59$ ). The models exhibit good discriminative ability between groups, with concordance 0.80 for extinction and 0.85 for mining. These results are consistent with larger worlds sustaining larger populations, reducing the probability of reaching the absorbing zero-population state while simultaneously exploring a larger phenotype space through more mutation events than smaller worlds.

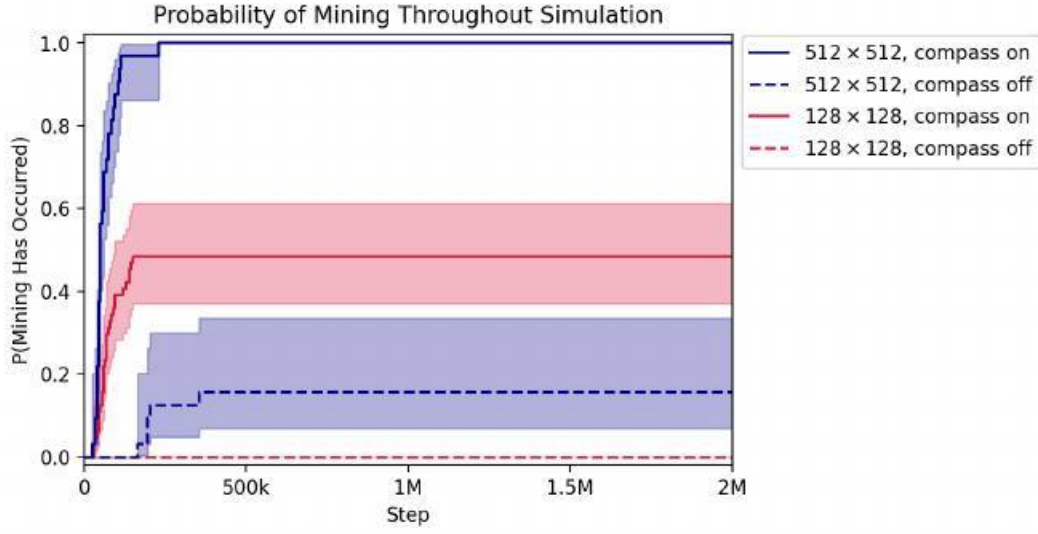


Figure 8: Estimated probability that mining has emerged as a function of world size, compass access, and timestep. We fit a separate non-parametric Kaplan-Meier estimator for each run group (e.g.  $512 \times 512$ , compass on). Shaded regions denote 95% confidence intervals. Each  $512 \times 512$  group fits the estimator on 32 runs; each  $128 \times 128$  group fits on 64 runs.

Table 7: Cox proportional hazards models for extinction and emergence events using a  $128 \times 128$ , compass-off baseline. Hazard ratios (HR) below 1 indicate reduced hazard and therefore longer time to event; for example, extinction  $HR = 0.04$  for  $512 \times 512$  means extinction occurs at 0.04 times the  $128 \times 128$  rate. The analysis includes 192 total runs: 32 seeds with compass on and 32 seeds with compass off in the  $512 \times 512$  world, and 64 seeds with compass on and 64 seeds with compass off in the  $128 \times 128$  world.

| Covariate             | Extinction |           |         | Mining   |             |         |
|-----------------------|------------|-----------|---------|----------|-------------|---------|
|                       | HR         | 95% CI    | p-value | HR       | 95% CI      | p-value |
| Size $512 \times 512$ | 0.04       | 0.02–0.07 | < 0.005 | 6.07     | 3.61–10.20  | < 0.005 |
| Compass On            | 0.03       | 0.01–0.05 | < 0.005 | 34.59    | 13.51–88.57 | < 0.005 |
| <b>Concordance</b>    | 0.80       |           |         | 0.85     |             |         |
| <b>Events / Total</b> | 135 / 192  |           |         | 68 / 192 |             |         |

## C ADDITIONAL FORAGING AND PREDATION RESULTS

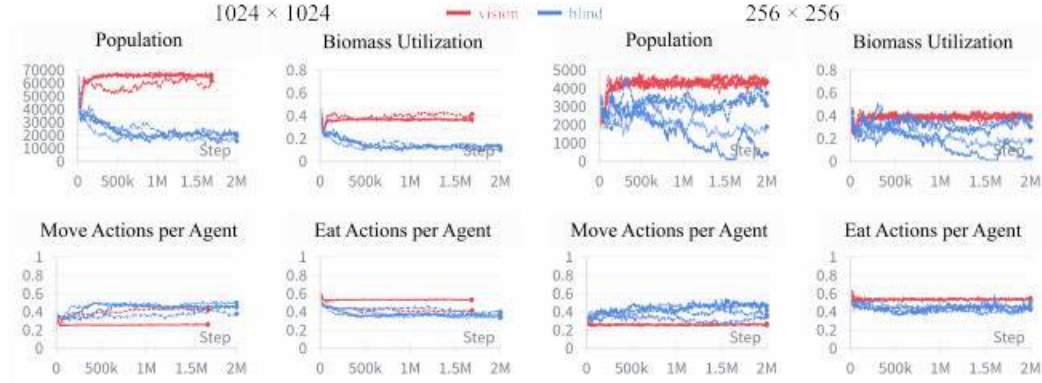


Figure 9: Foraging behavior in the **Ocean** world across scales. Enabling vision (red) produces more stable populations and higher biomass utilization through more frequent effective eating and reduced movement. At larger scales, these dynamics exhibit lower cross-seed variance. Due to GPU-hour limitations, the vision seeds at  $1024 \times 1024$  were run for 1.7M rather than 2M steps.

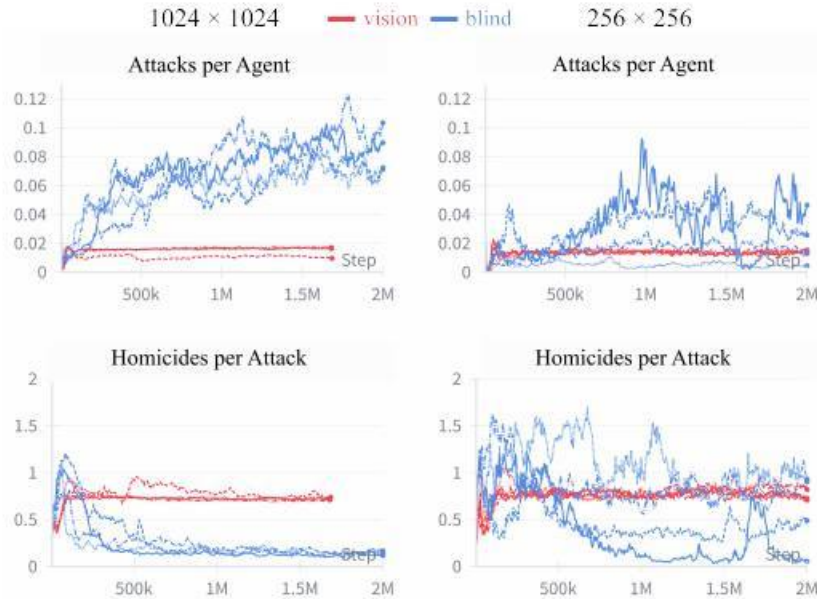


Figure 10: Predation behavior in the **Ocean** world across scales. With vision (red), agents attack less frequently but with substantially higher precision, consistent with specialized predation in addition to the foraging capabilities shown in Figure 9. The effect is stronger at larger scales, while blind agents (blue) exhibit higher behavioral variance in smaller worlds.

## D ADDITIONAL EXPERIMENT VISUALIZATIONS

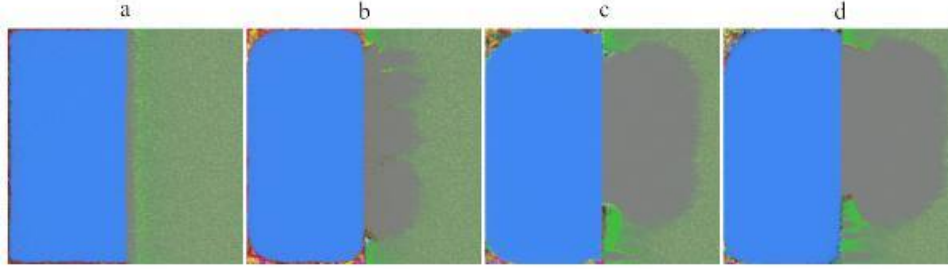


Figure 11: Large-scale environments induce more heterogeneous ecological dynamics and corresponding spatial patterns. In this  $1024 \times 1024$  **Beach** world, (**RC**) agents initially died near the shoreline (a), transferring biomass onto land. Many agents later evolved land-return behavior and mined this biomass (b) (c). After most land biomass had been mined, agents formed large clusters near the map boundary (d) (e).

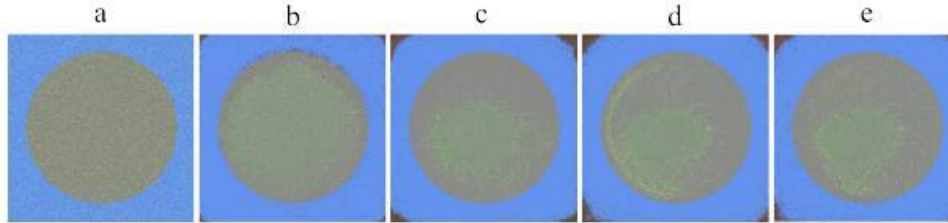


Figure 12: Mining behavior emerges across terrain geometries. In this  $512 \times 512$  **Island** world with (**RC**) agents, distinct shoreline regions are mined over the course of the run (b) (c). Biomass is returned to parts of the water (d) when agents die in transit, and the population remains stable through 2M steps.

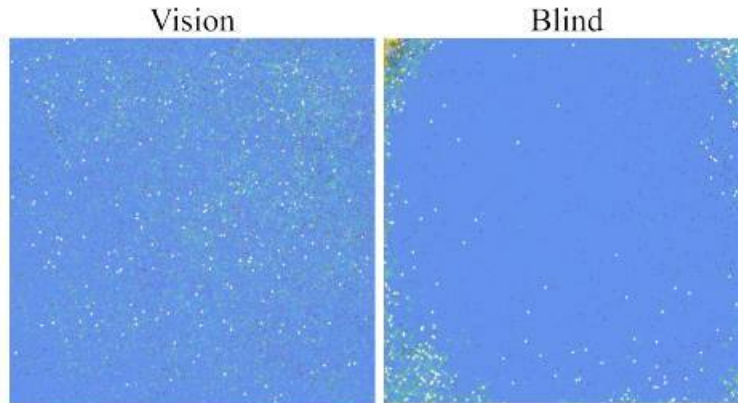


Figure 13: Representative  $512 \times 512$  **Ocean** runs with attacking enabled: visual (**RCV+A**) agents are shown on the left and blind (**RC+A**) agents on the right. White squares mark attacking agents. In the visual run, agents reliably target nearby agents, producing more effective predation, with quantitative comparisons reported in Figures 5 and 6. Blind agents sometimes reach high kill rates through the formation of dense clusters.

## E DETAILS ON COMPUTATION AND REPRODUCIBILITY

We provide additional details on simulator design, runtime scaling, and experiment reproducibility.

**Simulator design.** We conduct all experiments using a JAX-based, GPU-accelerated simulator designed for large-scale multi-agent ecosystems with open-ended populations. Our simulator computes every part of the environment step on the GPU, including terrain dynamics, resource flows, agent updates, and measurement, so that policy evaluation and simulation share a single JAX compilation pipeline. The simulator exposes a minimal population-level interface: given current map state, agent states, and heritable traits, a population algorithm supplies actions and updated traits and handles births, deaths, and environmental transitions. This separation allows the same environment implementation to be reused with different evolutionary algorithms, reinforcement learning methods, or other population update rules. All operations are implemented as pure JAX functions combined with vectorization and just-in-time compilation, enabling fusion into a small number of GPU kernels. This yields the throughput required for our scaling experiments, with runtime measurements for a single H100 GPU given in Table 8.

A key feature of the simulator is its built-in tooling, including a measurement API for computing and recording aggregate statistics such as biomass utilization, movement and eating rates, attack and homicide rates, and extensive environment-level parameters at each step, which we use directly in our figures and tables. In addition, an interactive 3D viewer renders terrain, water, resources, and agents from generated reports, and supports inspection of individual agents (Figure 1). These tools were critical for discovering and validating qualitative phenomena such as long-range resource mining and predation patterns that are not obvious from scalar metrics alone.

From a high-performance computing perspective, our simulator fills a gap relative to existing simulators. General-purpose agent-based frameworks such as MASON (Luke et al. 2005), FLAME-GPU (Richmond et al. 2024), or Griddly (Bamford et al. 2020) offer high configurability but require substantial environment engineering and do not directly target the open-ended, evolving populations at the scales we study. Conversely, GPU-native multi-agent RL environments emphasize fixed numbers of agents and task-specific reward structures. We combine GPU-native execution, explicit support for births and deaths with mutable traits, and configurable resource and terrain dynamics, enabling open-ended populations of over  $10^5$  agents while remaining interoperable with diverse population algorithms. This combination yields a state-of-the-art platform for scaling ecological experiments like those in this paper and a useful foundation for future work on emergent behavior in large populations. Our simulator will be released upon acceptance.

**Runtime data.** Table 8 reports runtime as a function of grid size and maximum agent population. Each run was executed on a single H100 GPU for at most 2M steps. The table reports runtimes for (RC) and (RCV + A) agents with 2 MLP layers and 64 channels per layer ( $\sim 22k$  parameters total); runtime also depends on network size, enabled sensors, and logging configuration.

At  $1024 \times 1024$  resolution, the largest runs require approximately  $\sim 10$  hours for the **Beach** terrain and  $\sim 14$  hours for the **Ocean** terrain on one H100, with memory usage of roughly 60GB. **Ocean** runs are slower because they enable both attacking and visual sensing.

Table 8: Runtime measurements for **Beach** and **Ocean** worlds across grid sizes and maximum agent populations. Each (RC) agent has a 22,465-parameter neural network, and each (RCV + A) agent has a 22,529-parameter neural network. The table reports simulation steps per second (hz) and extrapolated wall-clock runtime in hours for 2M steps. Effective runtime is lower for seeds in which the population goes extinct before 2M steps.

| Environment     | $1024 \times 1024$ |          | $512 \times 512$ |         | $256 \times 256$ |         | $128 \times 128$ |         | $64 \times 64$ |         |
|-----------------|--------------------|----------|------------------|---------|------------------|---------|------------------|---------|----------------|---------|
|                 | hz                 | runtime  | hz               | runtime | hz               | runtime | hz               | runtime | hz             | runtime |
| Beach (RC)      | 55                 | 10.1 hrs | 190              | 2.9 hrs | 315              | 1.8 hrs | 535              | 1.0 hrs | 750            | 0.7 hrs |
| Ocean (RCV + A) | 40                 | 13.9 hrs | 142              | 3.9 hrs | 235              | 2.4 hrs | 415              | 1.3 hrs | 750            | 0.7 hrs |

**Reproducibility.** Because the environments are large and stochastic, precise experimental outcomes can vary substantially across runs. We therefore report four random seeds for each experiment. Repeating an experiment with the same random seed does not reliably reproduce an identical trajectory because some low-level GPU operations are nondeterministic. In particular, modern GPUs may execute parallel floating-point operations, such as reductions, in different orders across runs and may

fuse operations in ways that alter rounding. Small finite-precision differences can accumulate over many simulation steps into observable trajectory drift. The use of 16-bit weights can amplify rounding coarseness, but it is not the primary source of nondeterminism. Despite run-level differences, clear aggregate patterns are visible across multiple seeds.

## F ABLATIONS

We report ablations that quantify sensitivity to selected hyperparameters and assess the robustness of the main results under alternative environment and agent configurations.

### F.1 INITIAL POPULATION SIZE SWEEP FOR FIXED MAP SIZE

We evaluate how initial population size affects mining consistency and population stability in a  $256 \times 256$  **Beach** world. Results are reported for **(RC)** populations. Agents without a compass **(R)** exhibited a 0% mining rate and a 100% extinction rate for all initial population sizes except 8192, the largest value tested, where mining occurred in 25% of seeds and extinction occurred in 75%. Mining and extinction frequencies are not mutually exclusive: a population may develop mining and later go extinct, or may persist for the full 2M-step horizon without developing mining.

Figure 14 indicates that initial population size has no substantial effect on mining consistency or extinction frequency: populations initialized with 128 agents evolve mining as often as populations initialized with 8192 agents, and extinction rates remain largely invariant across the sweep.

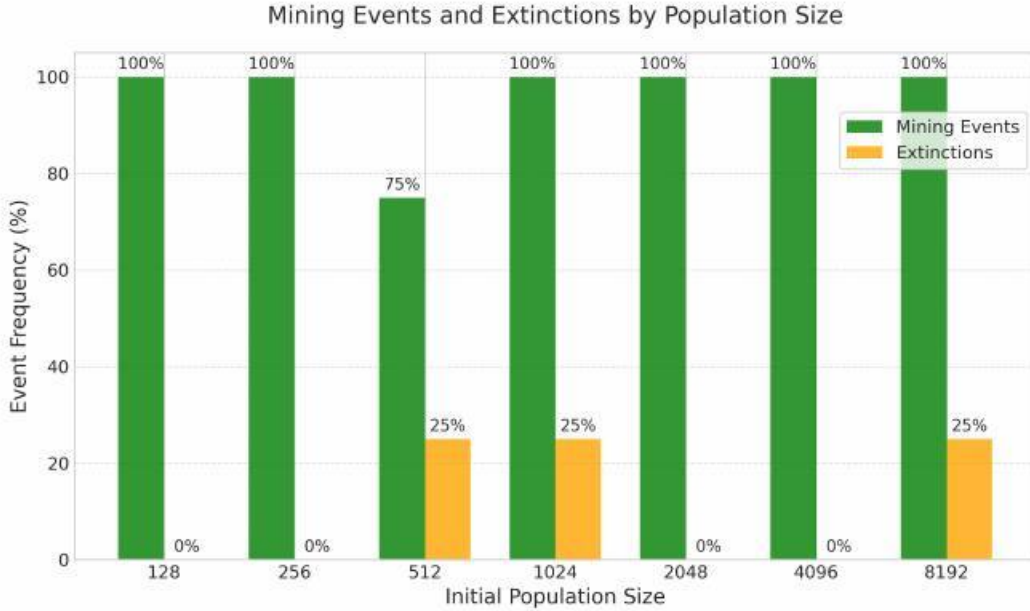


Figure 14: Frequency of mining and extinction events in a  $256 \times 256$  **Beach** world as a function of initial **(RC)** population size. Frequencies are computed over four random seeds for each initial population size; for example, a 25% Extinction Event Frequency indicates that 1/4 seeds reached population zero before 2M simulation steps.

Figure 15 explains why the observed effect is small. Irrespective of initial population size, each run rapidly approaches a similar population level during the first 50,000 simulation steps. This pattern suggests that the fixed resource budget induces an implicit environmental carrying capacity: populations initialized below this capacity grow toward it, whereas populations initialized above it are constrained downward. Consequently, effective population size becomes similar after an early transient, leaving mining and extinction frequencies largely unchanged. Larger initial populations nevertheless add some resources to the system because agents carry biomass that is deposited at death; therefore, larger initial populations reach slightly higher average maximum populations over

the full simulation (Table 9), although not enough to change mining or extinction frequency in this setting. After the early convergence phase, population trajectories may diverge substantially across seeds because different evolved abilities alter later resource distributions.

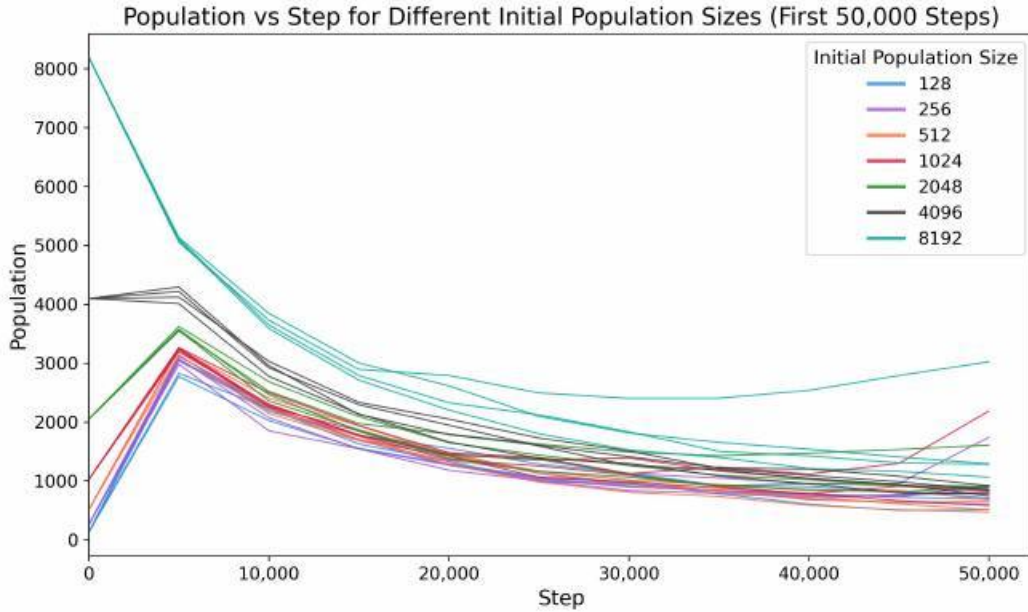


Figure 15: Population size during the first 50,000 time steps in the  $256 \times 256$  **Beach** with (RC) agents, using four seeds per initial population size. The environment induces an implicit resource-dependent carrying capacity, and all initial population sizes approach this value early in the run. Larger initial populations add biomass through agent death, which modestly increases the effective carrying capacity.

Table 9: Maximum post-initialization population size for each starting population size, reported as mean and standard deviation over four random seeds. Larger initial populations reach slightly higher maximum populations and exhibit greater cross-seed variance. However, the maximum population is largely robust to initialization: an initial population of 128, which is 1.6% of 8192, reaches 78.9% of the maximum population reached by the 8192 condition (3178 vs. 4026).

| Initial Population Size | Max Population Size After Step 0 |
|-------------------------|----------------------------------|
| 128                     | $3178 \pm 217$                   |
| 256                     | $3056 \pm 61$                    |
| 512                     | $3451 \pm 437$                   |
| 1024                    | $3613 \pm 381$                   |
| 2048                    | $3786 \pm 431$                   |
| 4096                    | $3689 \pm 437$                   |
| 8192                    | $4026 \pm 642$                   |

## F.2 VARYING RESOURCE RULES

We evaluate how two resource hyperparameters, biomass photosynthesis rate and initial biomass site density, affect mining and extinction frequency. The setting matches Appendix F.1: the initial population is fixed at 2048 agents, and all agents are compass-equipped (RC).

The biomass photosynthesis rate determines how quickly landscape biomass creates energy through photosynthesis, thereby supplying resources over time. The initial biomass site density specifies the

fraction of grid cells containing biomass at simulation start. All main-text experiments (Section 4) use a biomass photosynthesis rate of 0.01 and an initial biomass site density of 1/4.

Although initial population size did not materially change mining or extinction consistency (Figure 14), modifying resource rules produces a substantial effect, as shown in Figures 16 and 17

Mechanistically, increasing either biomass photosynthesis rate or initial biomass site density raises the effective carrying capacity by increasing available resources. Larger supported populations increase the probability of mutations that enable mining while reducing extinction risk. The data are consistent with this interpretation: excluding extinct seeds, runs with initial biomass site density 1/16 reached a mean post-step-0 maximum population of 1608, compared with 2466 for density 1/8, 3786 for density 1/4, and 5353 for density 1/2.

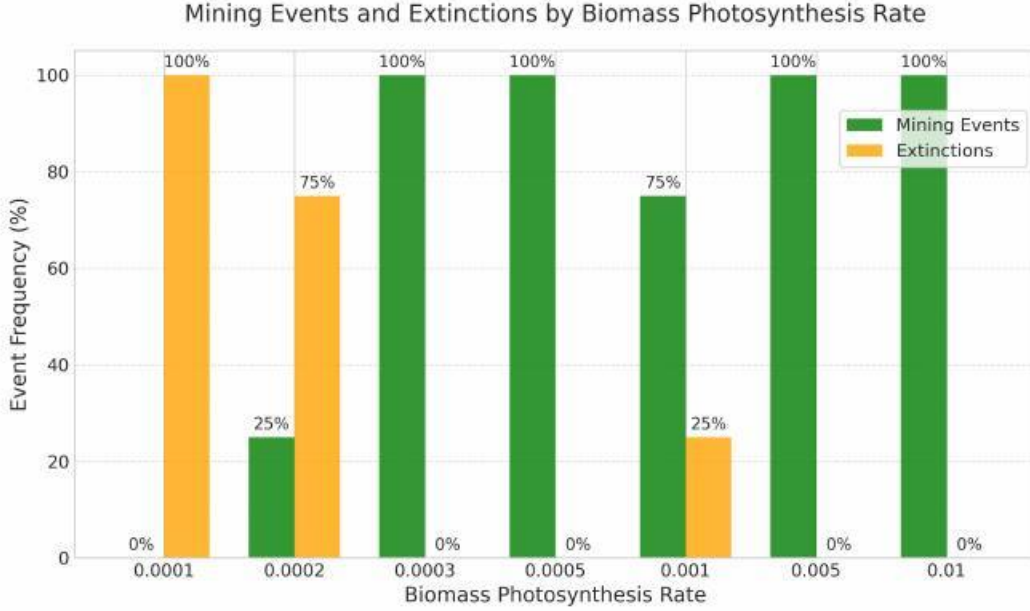


Figure 16: Frequency of mining and extinction events in a  $256 \times 256$  **Beach** world as a function of biomass photosynthesis rate. Frequencies are computed over four random seeds per setting. Mining emerges consistently at biomass photosynthesis rates of 0.0003 and greater.

### F.3 RULING OUT SELF-AVERAGING IN LARGER POPULATIONS

One possible explanation for the increased stability of mining in large environments (Table 1) is self-averaging rather than a qualitative change in dynamics. Here, self-averaging refers to the statistical-physics intuition that, as system size grows, fluctuations in macroscopic quantities shrink approximately as  $1/\sqrt{N}$ , where  $N$  denotes system size, so that a single large system behaves like the average of smaller systems with the same total size. Under this interpretation, a  $512 \times 512$  **Beach** world would function like many weakly coupled  $128 \times 128$  worlds executed in parallel. If so, the scaling results could in principle be reproduced by running sufficiently many small worlds and averaging over seeds.

To test this hypothesis, we approximately hold total “sample size” fixed while varying its partition across environments. In the **Beach** terrain with compass-equipped agents (**RC**), we evaluate three conditions:

- 4 runs of a  $512 \times 512$  world with 8192 initial agents each,
- 16 runs of a  $256 \times 256$  world with 2048 initial agents each,
- 64 runs of a  $128 \times 128$  world with 512 initial agents each.

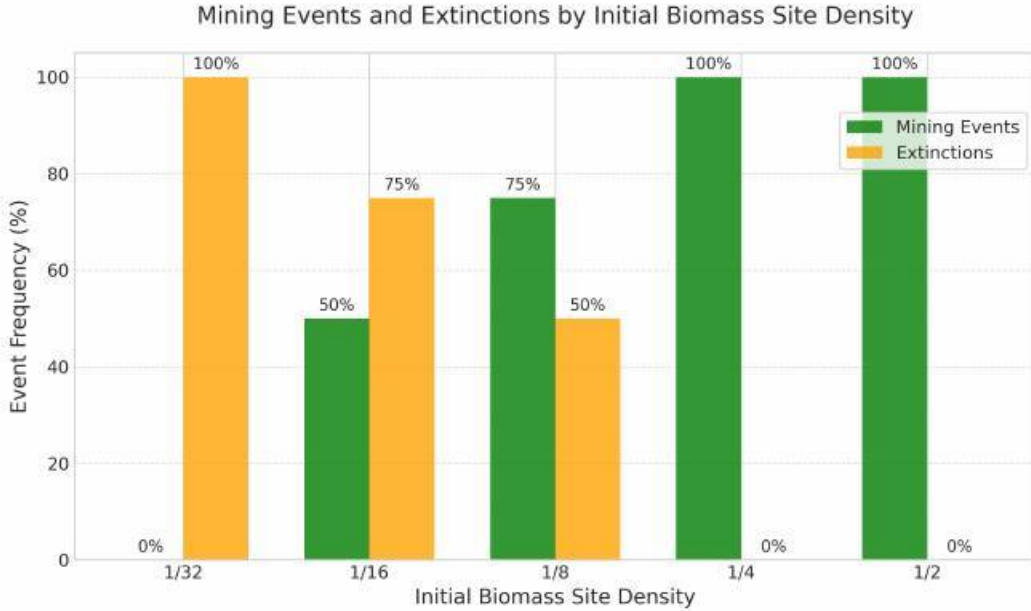


Figure 17: Frequency of mining and extinction events in a  $256 \times 256$  **Beach** world as a function of initial biomass site density. Frequencies are computed over four random seeds per setting. Higher initial biomass site density yields more consistent mining discovery and lower extinction probability.

Across these conditions, the aggregate number of grid cells, total starting resources, and total number of initial agents are identical:

$$4 \times 512^2 = 16 \times 256^2 = 64 \times 128^2, \quad 4 \times 8192 = 16 \times 2048 = 64 \times 512.$$

Each seed is run for 2M steps using the hyperparameters from Section 4.1. As a control, we repeat the protocol for agents without a compass (**R**). Table 10 summarizes the frequency of long-distance mining and extinction in each configuration.

If large environments were merely self-averages of smaller ones, mining and extinction rates should be approximately invariant to how grid cells and agents are partitioned across runs: four large worlds and sixty-four small worlds should exhibit similar mining-discovery probabilities. Instead, event frequencies depend strongly on world size. For (**RC**) agents, mining emerges in every  $512 \times 512$  run with no extinctions, remains common but not guaranteed at  $256 \times 256$ , and becomes unreliable at  $128 \times 128$ , where many populations go extinct before 2M steps. In contrast, (**R**) agents never develop mining and go extinct at all scales, confirming that compass information remains a key enabling variable for long-range behavior.

These results are inconsistent with a pure self-averaging account. Larger environments appear to enable and stabilize mining through mechanisms that are not reproduced by averaging over additional small simulations.

Even when total cells, agents, and simulation steps are matched across conditions, concentrating resources into a small number of large worlds yields consistent mining and rare extinction, whereas distributing the same resources across many small worlds yields frequent mining failures and higher collapse rates. This suggests that increased environmental scale does more than smooth stochastic noise: it changes the ecological dynamics, for example by supporting larger, more spatially distributed populations and more persistent policy diversity.

#### F.4 MUTATION RATE SWEEP

We test the sensitivity of mining emergence to the mutation-rate hyperparameter, which sets the scale of Gaussian noise added to an offspring policy at birth. The mutation rate therefore controls

Table 10: Mining and extinction events under a tradeoff between world size and number of independent runs. We compare **Beach** worlds at  $512 \times 512$ ,  $256 \times 256$ , and  $128 \times 128$ , with initial populations scaled proportionally (8192, 2048, and 512 agents, respectively). For each size, the aggregate number of grid cells and initial agents is held fixed by running 4 seeds at  $512 \times 512$ , 16 seeds at  $256 \times 256$ , and 64 seeds at  $128 \times 128$ . The table reports the fraction of runs with mining emergence (🔧) and the fraction with extinction before 2M steps (💀), for agents with a compass (**RC**) and without a compass (**R**).

|      | $512 \times 512$<br>8192 |     | $256 \times 256$<br>2048 |       | $128 \times 128$<br>512 |       |
|------|--------------------------|-----|--------------------------|-------|-------------------------|-------|
|      | 🔧                        | 💀   | 🔧                        | 💀     | 🔧                       | 💀     |
| (RC) | 4/4                      | 0/4 | 15/16                    | 1/16  | 31/64                   | 41/64 |
| (R)  | 0/4                      | 4/4 | 0/16                     | 16/16 | 0/64                    | 64/64 |

the expected parameter-level deviation between child and parent. We again use a  $256 \times 256$  **Beach** world initialized with 2048 (**RC**) agents and run for at most 2M steps.

Figure 18 shows that for mutation rates from  $6 \times 10^{-3}$  through  $7.875 \times 10^{-2}$ , mining emerges in at least 3 of 4 seeds, indicating that the main-result mutation rate of  $3 \times 10^{-2}$  in Section 4.1 lies within a robust interval. Very small mutation rates, such as  $3 \times 10^{-3}$ , limit behavioral innovation and impair survival, leading to extinction. Conversely, very large mutation rates, such as  $3 \times 10^{-1}$ , destabilize the population and consistently produce extinction without mining adaptation.

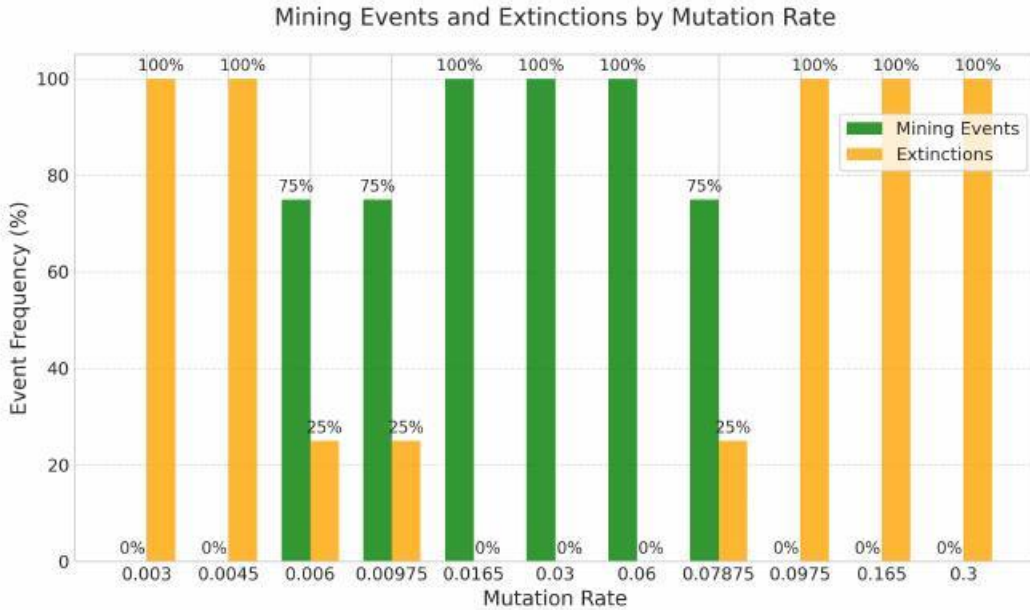


Figure 18: Frequency of mining and extinction events in a  $256 \times 256$  **Beach** world across mutation rates. Frequencies are computed over four random seeds for each setting.

## F.5 NETWORK SIZE

We analyze how policy-network size and architecture affect mining and extinction frequency, and how this dependence changes with environmental scale. We evaluate  $256 \times 256$  and  $512 \times 512$  **Beach** worlds initialized with 2048 and 8192 (**RC**) agents, respectively, preserving the fourfold area scaling. We also run (**R**) baselines for each condition and again observe rare to no mining events with 100% extinction rates.

The main experiments in Section 4 use policies with 2 layers and 64 channels, corresponding to 22465 parameters per agent (Table 11). Figure 19 shows that, in the  $256 \times 256$  **Beach**, reducing width to 32 channels (9185 parameters) or increasing width to 128 channels (61313 parameters) while holding depth fixed has little effect on mining or extinction frequency. By contrast, changing depth has a substantial effect: all populations with 1-layer, 32-channel policies went extinct, and 50% of populations with 3-layer, 128-channel policies went extinct.

Scaling the environment to  $512 \times 512$  changes these depth-dependent dynamics (Figure 20). In the larger world, populations with 3-layer, 128-channel networks consistently mined and survived for 2M steps. Populations with the smaller 1-layer, 32-channel networks also mined consistently. We hypothesize that small and large networks make behavioral discovery difficult for different reasons. Low-capacity networks may be unable to represent effective policies, whereas larger networks increase the dimensionality of the mutation search space and may require more time to reach useful behaviors. The observation that both architectures frequently fail in small environments but succeed in larger ones further illustrates the stabilizing effect of scale. Larger environments support larger populations, increasing exploration in policy space. At the same time, locally successful mutants propagate more slowly across a large spatial domain, preserving diversity and further increasing exploration. Related spatial coevolutionary effects are discussed in (Mitchell et al. 2006).

Table 11: Total parameter counts for (RC) and (R) agent policies under each network architecture.

| Architecture<br>(layers, channels) | Parameter Count<br>(RC) | (R)   |
|------------------------------------|-------------------------|-------|
| 1, 32                              | 8129                    | 7969  |
| 2, 32                              | 9185                    | 9025  |
| 2, 64                              | 22465                   | 22145 |
| 2, 128                             | 61313                   | 60673 |
| 3, 128                             | 77825                   | 77185 |

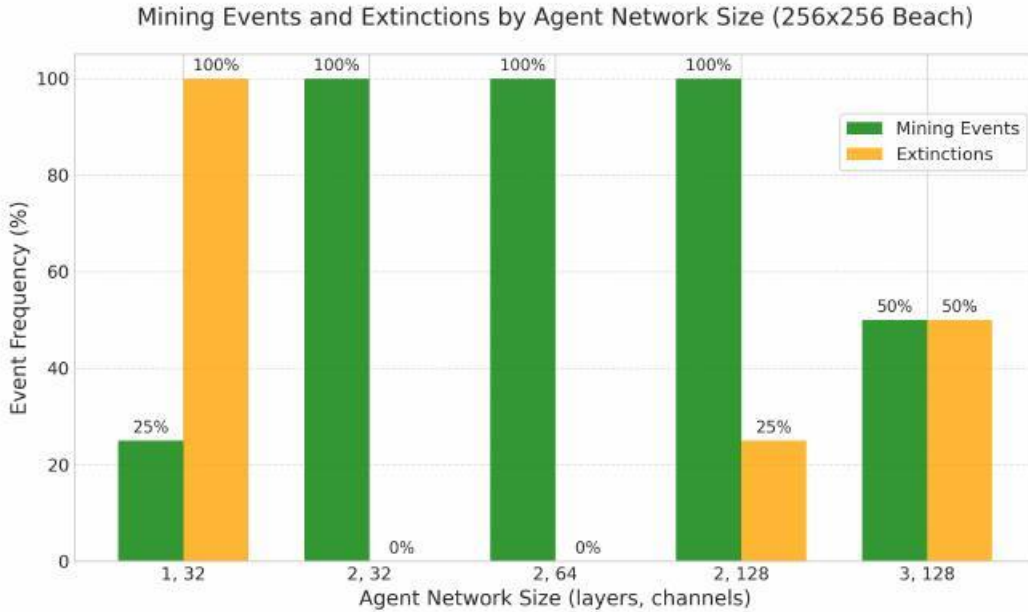


Figure 19: Frequency of mining and extinction events in a  $256 \times 256$  **Beach** world across agent policy network sizes. Frequencies are computed over four random seeds for each setting.

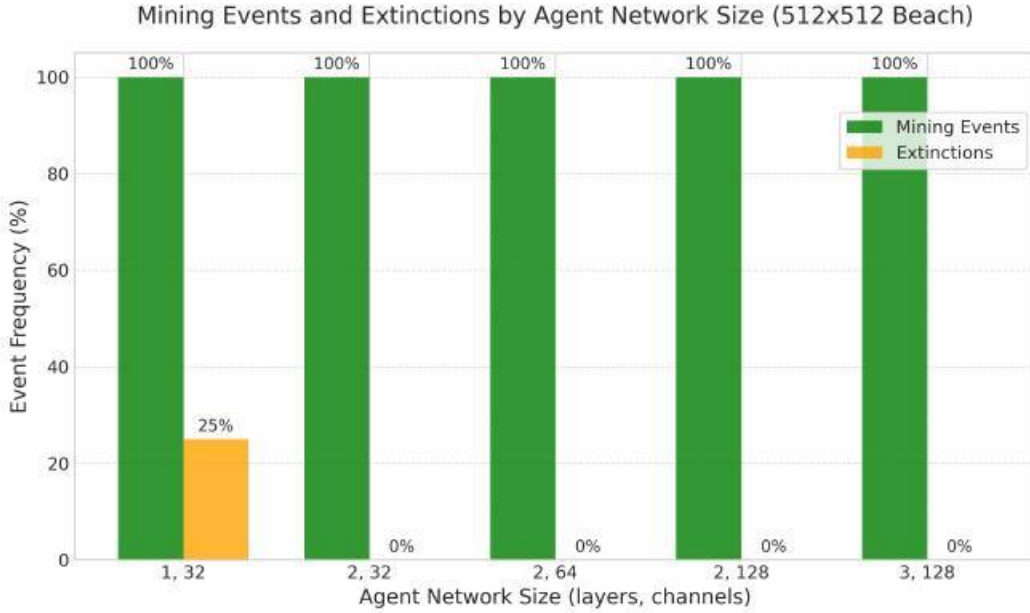


Figure 20: Frequency of mining and extinction events in a  $512 \times 512$  **Beach** world across agent policy network sizes. Frequencies are computed over four random seeds for each setting.

#### F.6 VARYING VISION RANGE

We assess how vision range and viewing geometry affect predation and foraging in the  $512 \times 512$  **Ocean** world with (RCV+A) agents. We evaluate visual sensor ranges of  $3 \times 3$ ,  $7 \times 7$ , and  $9 \times 9$ . For each range, we compare the two configurations in Figure 21 (a) Around, in which agents observe cells surrounding them, and (b) Front, in which agents do not observe cells behind them. The number of observed cells is identical across the two configurations. The main visual experiments in Section 4.2 use the  $7 \times 7$  Around setting.

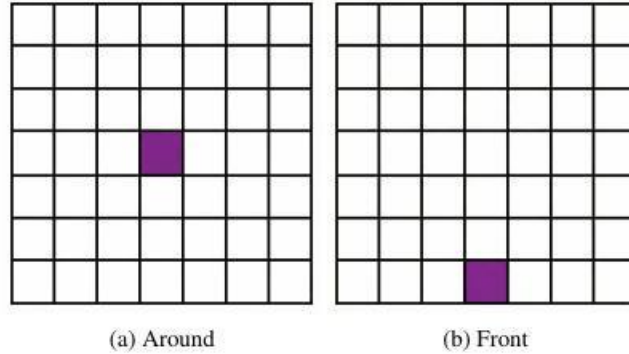


Figure 21: Comparison of the Around and Front viewing configurations for the  $7 \times 7$  vision range. The purple cell denotes an agent oriented toward the top of the page.

These results indicate that the smaller  $3 \times 3$  vision range produces behavior close to the Blind condition, whereas  $9 \times 9$  performance is similar to  $7 \times 7$ . Thus, in the present environment, increasing vision range exhibits diminishing returns. This pattern is consistent with the memoryless policy architecture: peripheral features in a  $9 \times 9$  window are unlikely to change the locally optimal action relative to a  $7 \times 7$  window often enough to provide a strong selection signal.

The data further suggest that at  $7 \times 7$  and  $9 \times 9$ , the Around configuration is slightly more favorable for population stability and the effectiveness of attack and eat actions, yielding higher average

Table 12: Foraging and predation measurements in the  $512 \times 512$  **Ocean** world for visual agents (**RCV+A**) under different vision ranges and viewing configurations. Tables 5 and 6 define the icon-coded metrics. For each seed, each metric is averaged over steps beginning at step 500k to allow transient dynamics to stabilize. Reported values are the mean across four seeds and the standard deviation of the four seed-level averages.

| Vision Range | Viewing Configuration |  |  |  |  |  |  |
|--------------|-----------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| $3 \times 3$ | Blind                 | $7212 \pm 2496$                                                                   | $.181 \pm .060$                                                                   | $.435 \pm .090$                                                                   | $.392 \pm .033$                                                                     | $.032 \pm .006$                                                                     | $.307 \pm .090$                                                                     |
|              | Around                | $10308 \pm 4016$                                                                  | $.258 \pm .105$                                                                   | $.352 \pm .014$                                                                   | $.427 \pm .022$                                                                     | $.040 \pm .030$                                                                     | $.388 \pm .195$                                                                     |
|              | Front                 | $13863 \pm 520$                                                                   | $.351 \pm .023$                                                                   | $.336 \pm .008$                                                                   | $.437 \pm .009$                                                                     | $.026 \pm .006$                                                                     | $.470 \pm .145$                                                                     |
|              | Around                | $16739 \pm 122$                                                                   | $.373 \pm .003$                                                                   | $.259 \pm .000$                                                                   | $.532 \pm .001$                                                                     | $.015 \pm .000$                                                                     | $.751 \pm .007$                                                                     |
|              | Front                 | $15255 \pm 623$                                                                   | $.343 \pm .015$                                                                   | $.253 \pm .001$                                                                   | $.540 \pm .004$                                                                     | $.015 \pm .000$                                                                     | $.690 \pm .041$                                                                     |
|              | Around                | $16744 \pm 117$                                                                   | $.372 \pm .002$                                                                   | $.257 \pm .001$                                                                   | $.534 \pm .001$                                                                     | $.015 \pm .000$                                                                     | $.728 \pm .009$                                                                     |
| $9 \times 9$ | Front                 | $15211 \pm 1416$                                                                  | $.342 \pm .033$                                                                   | $.253 \pm .001$                                                                   | $.541 \pm .007$                                                                     | $.015 \pm .000$                                                                     | $.664 \pm .050$                                                                     |

population , biomass utilization , and homicides per attack  than the Front configuration. We hypothesize that this reflects diminishing returns from observing distant pixels: Around allocates more observed cells to nearby locations, which are more likely to contain action-relevant information than the distant cells visible in Front. At  $3 \times 3$ , the trend reverses, with Front outperforming Around; in this environment, the marginal value of observing one additional forward pixel in the  $3 \times 3$  Front configuration appears to exceed the value of observing one rear pixel in the  $3 \times 3$  Around configuration, which behaves most similarly to Blind.

## F.7 VARYING ATTACK STRENGTH

We study how reducing attack strength affects emergent behavior and population dynamics in the  $256 \times 256$  **Ocean** world. We evaluate three settings: 1) Full Attack, which deals 10 HP of damage and instantly kills any agent in the attack radius; 2) Half Attack, which deals 5 HP of damage; and 3) Quarter Attack, which deals 2.5 HP of damage. The main **Ocean** experiments in Section 4.2 use Full Attack.

Reducing attack strength from Full Attack to Half Attack substantially decreases both attack frequency and attack effectiveness: attacks per agent and homicides per attack both decline (Figure 22). Consequently, lower-attack-strength populations become larger, and agents move and eat more frequently, increasing biomass utilization. Further reducing strength to Quarter Attack yields diminishing additional reductions in violence and diminishing population-growth gains. Mechanistically, lower damage means that a homicide requires more attacks, reducing per-attack lethality. Once lethality decreases, attacking becomes less selectively useful, so the population evolves less specialized attacking behavior and allocates more behavior to foraging.

## F.8 VARYING AGENT HP

We investigate whether reducing offspring starting HP and lowering the healing rate imposes a lifetime growth pressure on agents. The healing-rate parameter determines how much HP an agent regenerates per healing step by converting available energy into HP. Agents can affect healing only indirectly by maintaining sufficient energy reserves.

We evaluate two settings: 1) Full HP, in which child HP starts at 5 and the healing rate is 1 HP per 0.1 energy; and 2) Low HP, in which child HP starts at 1 and the healing rate is 0.5. Maximum agent HP is 10 in all settings. The main experiments (Section 4) use Full HP.

Across the two settings, homicides per attack, eat-action frequency, and biomass utilization show no substantial differences (Figure 23). Full HP produces slightly larger populations, whereas Low HP produces lower attack frequency and slightly higher movement. We hypothesize that low-HP offspring must prioritize early-life foraging to reach full size, which could account for this behavioral shift.

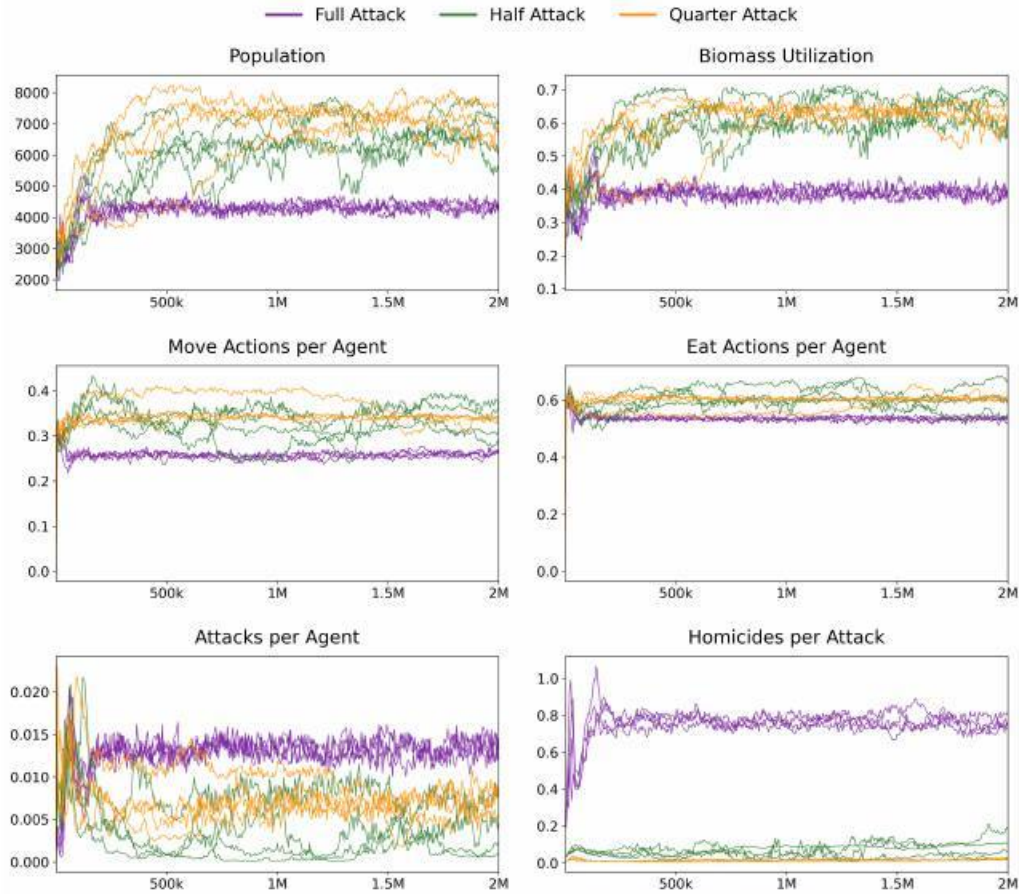


Figure 22: Population and action-frequency measurements over 2M steps in a  $256 \times 256$  Ocean world for (RCV+A) agents under Full Attack, Half Attack, and Quarter Attack settings. Four seeds are plotted for each attack-strength setting.

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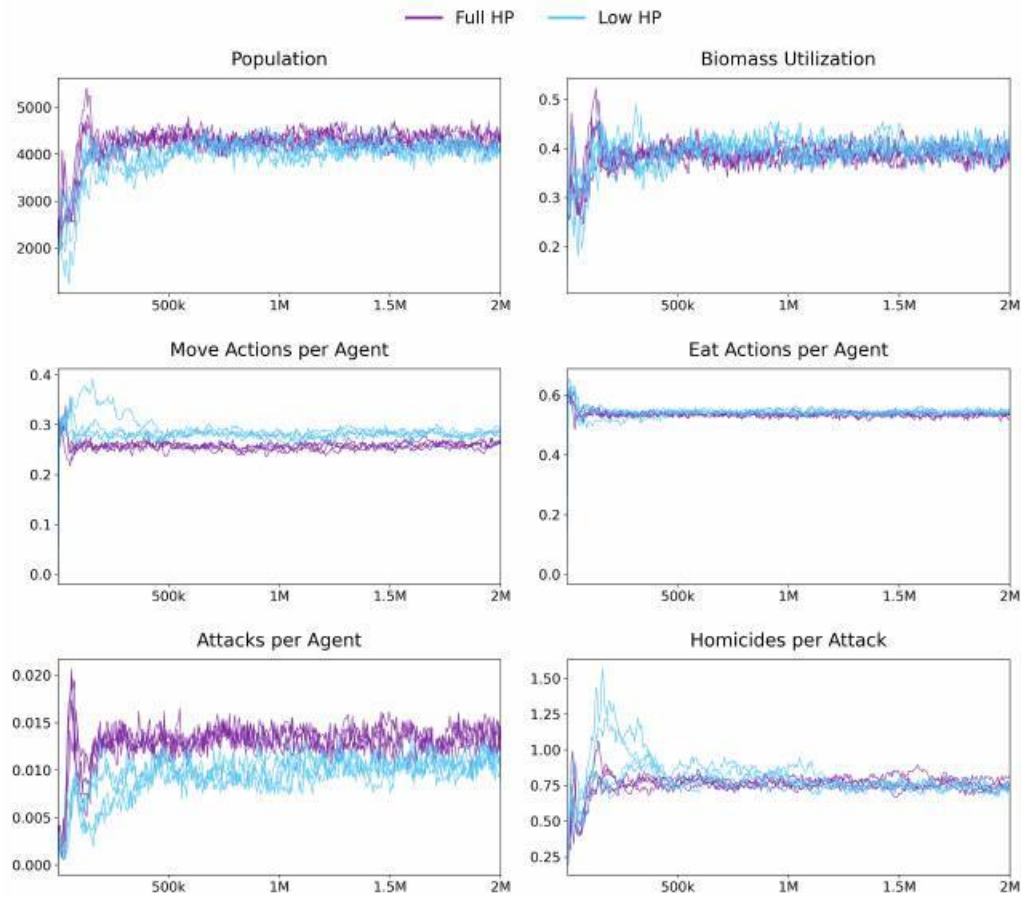


Figure 23: Population and action-frequency measurements over 2M steps in the  $256 \times 256$  Ocean world for (RCV+A) agents under Full HP and Low HP settings. Four seeds are plotted for each HP setting.